PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

Condominium Booking and Operations Management System

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

The integration of information technology into business operations has become a critical factor in improving organizational efficiency, data accuracy, and service quality across industries [1]. As digital transformation has accelerated in every sector of the global economy, businesses have increasingly adopted computerized systems to streamline daily operations, minimize human error, and establish more reliable data management infrastructures [2]. Organizations that fail to embrace technological advancement struggle to compete with digitally equipped counterparts, particularly as client expectations rise and regulatory requirements for accurate record-keeping intensify [3].

In the real estate and property management industry, digital transformation plays an especially vital role. Managing bookings, processing financial transactions, maintaining tenant records, and scheduling property maintenance are inherently complex activities that become unmanageable at scale without automated systems [4]. Property managers overseeing multiple units across different buildings or locations face compounding administrative burdens when relying on manual processes. These burdens result in delayed responses to tenant concerns, inaccurate financial reporting, and missed business opportunities. The adoption of integrated property management platforms has therefore shifted from a competitive advantage to an operational necessity [5].

The growing demand for condominium rentals in urban areas amplifies the need for efficient property management solutions. This demand is driven by rapid urbanization, increased migration of students and young professionals to city centers, and shifting preferences toward high-density, amenity-rich residential environments [6]. Condominium rentals offer distinct advantages over traditional house rentals, including enhanced security features, proximity to commercial and educational institutions, shared facility access, and predictable utility arrangements. These qualities make condominium units the preferred housing choice for a

growing demographic of urban residents. In turn, property management businesses face greater pressure to maintain seamless operations and high standards of tenant service.

Managing multiple rental units simultaneously nonetheless introduces considerable administrative complexity. Staff must track tenant inquiries across different communication channels, maintain accurate booking schedules to prevent conflicts, monitor payment statuses for tenants with varying payment cycles, and respond promptly to maintenance requests. These tasks require systems capable of processing, organizing, and retrieving large volumes of data in real time. Without such systems, administrative staff must devote disproportionate amounts of time to clerical tasks. This reduces their capacity for higher-value activities that directly improve tenant satisfaction and business performance [7].

1.2 Statement of the Problem

Despite its growing clientele and expanding portfolio of rental units, Gales Urban Haven Unit relies predominantly on manual processes for its core operational functions:

1. Booking management. Booking records are kept in handwritten logs or basic spreadsheet entries, with no automated mechanism to verify unit availability or flag scheduling conflicts.
2. Payment tracking. Staff manually reconcile tenant payment receipts against ledger records, a process that becomes increasingly error-prone as transaction volume grows.
3. Revenue computation. Spreadsheet-based calculations are vulnerable to formula errors, data entry inconsistencies, and version control issues when multiple users access shared files.
4. Maintenance monitoring. Maintenance activities are recorded informally through verbal communication, handwritten notes, or disconnected digital files. There is no centralized system to assign tasks, monitor progress, or verify resolution.

1.3 Objectives of the Study

1.3.1 General objective

The objective of this study is to design, develop, and implement a centralized Condominium Booking and Operations Management System for Gales Urban Haven Unit (One Spatial Iloilo). The system serves as a comprehensive digital solution that automates and integrates the core operational workflows of the business, namely booking management, payment monitoring, revenue tracking, calendar-based scheduling, and maintenance recording. It replaces the current manual and fragmented processes with a secure, reliable, web-based platform. The platform is intended to improve operational efficiency, minimize human error, enhance data accessibility, and support faster and more accurate business decisions. By centralizing operational data in a single platform, the system gives administrators and property managers greater confidence,

control, and precision in daily operations. It also establishes a scalable digital infrastructure that can adapt to the growing demands of the business, sustain service quality, and optimize resource utilization.

1.3.2 Specific Objectives

1. **Analyze the existing booking and operational procedures.** The analysis identifies inefficiencies, recurring challenges, workflow bottlenecks, and the system requirements needed for automation. It examines how reservations are recorded, payments tracked, maintenance requests managed, and scheduling conflicts handled. Data is gathered through direct consultations with the business owners and administrative staff and through a review of current documentation, record-keeping methods, and communication workflows. The volume and frequency of bookings, payments, and maintenance requests are also assessed to determine the scale of the data management requirements. The findings establish clear, measurable requirements so that each feature responds to a validated operational need, and they guide the prioritization of features during development.
2. **Design a secure, accessible, and user-friendly web-based system.** The design follows user-centered principles and draws on established interface patterns and usability standards for web applications. It defines user roles and role-based access controls, applies secure authentication and encryption, and uses a responsive layout that works on desktops, laptops, and tablets. The interface is structured to minimize the learning curve for staff. Data validation and error-handling mechanisms guide users toward complete and accurate entries. The design also prioritizes scalability and performance under high usage and large data volumes.
3. **Develop an automated booking management module.** The module allows administrators to record and process reservation requests, check real-time unit availability, and prevent scheduling conflicts and double bookings. It maintains an up-to-date repository of active, pending, and completed reservations, sends notifications and confirmation alerts, and produces booking reports and analytics for monitoring occupancy trends and forecasting demand. A centralized dashboard gives administrators an organized overview of reservation activity.
4. **Implement a payment tracking module.** The module records and monitors tenant down payments, outstanding balances, payment history, and cumulative revenue in a structured and auditable manner. It provides a real-time view of the payment status of every unit, reduces discrepancies caused by manual computation, and keeps a verifiable trail of transactions. Automated reminders and alerts support follow-up on pending and overdue payments.

5. **Create a revenue reporting module.** The module automatically generates financial summaries, itemized payment reports, and revenue analytics on demand and at scheduled intervals. Income data are presented in visual and tabular form so that management can assess financial performance across periods. Automating report generation reduces the time spent on manual reporting and the risk of computational errors.
6. **Integrate a calendar-based scheduling system.** A unified, interactive calendar lets administrators monitor reservations, check-ins and check-outs, unit availability windows, and scheduled maintenance. It is synchronized with the booking and maintenance modules, so status changes are reflected system-wide. Features include color-coded event indicators, category filters, daily, weekly, and monthly views, and automated alerts for upcoming events. The calendar keeps a timestamped record of scheduled and completed activities to support operational review and planning. Administrators can add, modify, and remove entries with minimal effort, which removes the need for separate scheduling tools.
7. **Develop a maintenance monitoring module.** The module records, tracks, and manages repair requests, preventive maintenance schedules, and service completion updates. Administrators can log requests with detailed descriptions, assign tasks to personnel, set priority levels, and track progress from initiation to completion. Scheduled maintenance appears automatically in the operational calendar, and automated reminders flag upcoming preventive tasks and overdue requests. Records are timestamped and can be filtered by unit, maintenance type, priority level, and completion status. This supports property condition assessment, warranty tracking, pattern identification, and proactive maintenance planning, while giving personnel a transparent record of assignments and accountability.
8. **Establish a centralized database management system.** A single relational database securely stores booking records, payment transactions, tenant profiles, maintenance logs, and revenue reports. It is designed for data integrity, consistency, and security. It supports access controls, automated backups, validation constraints that prevent incomplete or duplicate records, and concurrent multi-user access. The database serves as the single source of truth for all system-generated reports and analytics. Its schema is scalable to accommodate growth in records, transactions, and user accounts.
9. **Evaluate the functionality, usability, and efficiency of the developed system.** Evaluation uses structured user acceptance testing with administrators, business owners, and other authorized users. It also uses system performance testing of response time, processing speed, and stability under varying loads, and structured feedback collection through questionnaires and rating scales. The evaluation also verifies the accuracy and consistency of system-generated reports and records. Issues identified are documented and addressed through targeted refinements, and the results are analyzed against the

study's objectives to determine how well the system resolves the identified operational inefficiencies.

1.4 Scope and Limitations

1.4.1 Scope of the Study

The study covers the design, development, and evaluation of a web-based Condominium Booking and Operations Management System for Gales Urban Haven Unit. The system includes the following modules:

- booking management, with real-time availability checking and double-booking prevention;
- payment tracking, covering down payments, balances, and payment history;
- revenue reporting and analytics;
- calendar-based scheduling of reservations, check-ins and check-outs, and maintenance activities;
- maintenance monitoring; and
- a centralized relational database with role-based access control.

The system is intended for use by administrators and property managers, and it is evaluated through user acceptance testing, performance testing, and feedback from end users and stakeholders.

1.4.2 Delimitations of the Study

Certain features were deliberately excluded to keep the project focused, feasible, and appropriate for an academic study. These delimitations reflect constraints in time, resources, and technical infrastructure. They let the researchers concentrate on the core functionalities with the most immediate operational benefit to Gales Urban Haven Unit (One Spatial Iloilo). They also mark natural areas for future development. Each is described below with its rationale.

- **External payment gateway and banking integration.** The system does not integrate with third-party payment gateways such as PayMaya, GCash, or PayPal, or with banking APIs. Such integrations require compliance with financial regulations, merchant account agreements, secure communication protocols, and ongoing API maintenance, all beyond the scope of an academic capstone project. The Payment Recording Module instead documents payments that have already been received, which is consistent with the client's current practice. It provides a foundation on which future developers can build online payment processing.

- **Native mobile application development.** The study does not include a native Android or iOS application. Mobile development is a distinct discipline with its own tools, frameworks, and platform requirements, and building it in parallel with the web system would exceed the available time and resources. The web interface is instead designed to be responsive where practicable, so it can be accessed from modern mobile browsers.
- **SMS and email notification services.** The system does not send SMS alerts or email confirmations to tenants or administrators. These features require third-party messaging platforms such as Twilio or SendGrid, or local telecommunications services. They also involve subscription costs, access agreements, and data privacy considerations under applicable regulations. The system handles notification-related information internally and gives administrators the data they need to contact tenants through their preferred external channels. Automated messaging is reserved for a future version.
- **Multi-property and enterprise-level management.** The system is designed for Gales Urban Haven as a single-property operation. It does not provide multi-property management, enterprise dashboards, or tools for administering multiple buildings or branches. Such systems typically need multi-tenant database architectures, hierarchical administrative structures, and extensive customization. The modular architecture, however, allows these features to be added in later versions.
- **Cloud hosting and advanced cybersecurity.** The study does not cover cloud-based hosting configurations, content delivery networks, server load balancing, or cybersecurity measures beyond standard authentication and basic database security. These require specialized infrastructure knowledge and resources beyond the scope of an academic study. The security architecture includes hashed password storage, session management, and basic input validation to guard against common vulnerabilities. These measures were appropriate for the academic and prototype context in which the system was deployed and evaluated.

1.5 Significance of the Study

The proposed Condominium Booking and Operations Management System was expected to provide benefits to various stakeholders by improving business processes and operational management.

• **Business Owner** – The study helped business owners improve operational efficiency through automated booking management, accurate payment monitoring, revenue tracking, and centralized record management. It also supported better decision-making through readily available reports and organized business data. Furthermore, business owners were able to monitor real-time updates on reservations and cancellations, allowing them to respond promptly to changes in demand. The system also provided detailed financial

summaries that enabled owners to identify trends, forecast future revenue, and allocate resources more effectively. In addition, business owners benefited from a reduced manual workload, as the automated processes minimized human errors and saved valuable time. The centralized platform also allowed owners to access and review all business records from a single location, which significantly improved overall business management and long-term planning.

• **Administrative Staff** – The system simplified daily operational tasks such as booking management, payment recording, schedule monitoring, and maintenance tracking. This reduced manual workload, minimized errors, and improved productivity. Additionally, the system provided administrative staff with a more organized and efficient workflow, allowing them to focus on higher-priority tasks rather than repetitive manual processes. Staff members were also able to retrieve and update records quickly, which eliminated unnecessary delays and ensured that all information remained accurate and up to date. The system further assisted administrative staff in generating reports and summaries, reducing the time spent on paperwork and manual data compilation. Moreover, real-time notifications and alerts kept the staff well-informed about upcoming bookings, pending payments, and scheduled maintenance, ensuring that operational concerns were addressed in a timely manner. As a result, administrative staff experienced a more manageable and less stressful work environment, ultimately contributing to a higher level of service quality and overall operational success.

• **Tenants and Clients** – Clients benefited from improved booking organization, reduced scheduling conflicts, and more efficient handling of reservations and maintenance concerns, resulting in better service quality and customer satisfaction. Furthermore, clients experienced a smoother and more convenient booking process, as the system allowed them to make and confirm reservations with greater ease and accuracy. They also received timely updates and notifications regarding their bookings, payments, and any changes to their reservations, ensuring that they remained well-informed. In addition, clients benefited from faster response times to maintenance concerns and service requests, as the system enabled staff to track and address issues more efficiently. The organized record management also ensured that client information and transaction history were stored securely and accurately, reducing the likelihood of disputes or miscommunication. Ultimately, the improved system fostered a more trustworthy and professional relationship between the business and its clients, leading to higher levels of loyalty and long-term customer satisfaction.

• **BSIT Students** – This study served as a practical application of concepts in web development, database management, software engineering, and systems analysis and design. It also provided students with experience in solving real-world business problems using information technology. Moreover, BSIT students gained hands-on experience in developing and implementing a fully functional system, which strengthened their

technical skills and deepened their understanding of the software development lifecycle. The study also exposed students to the challenges and complexities of building real-world applications, preparing them for future professional roles in the field of information technology. Additionally, students developed critical thinking and problem-solving abilities as they designed and refined system features to meet the specific needs of the business. The project also enhanced their ability to work collaboratively, manage project timelines, and communicate technical solutions effectively, which were valuable skills for their future careers. Furthermore, this study served as a significant addition to their academic portfolio, demonstrating their capability to create practical and impactful technology solutions that addressed genuine business needs.

• **Future Researchers** – The study served as a reference for future research related to property management systems, booking automation, revenue monitoring, and web-based business solutions. Researchers could use this study as a foundation for enhancing similar systems or exploring additional features and technologies. Furthermore, future researchers were able to build upon the findings and methodologies of this study to develop more advanced and comprehensive property management solutions. The documented processes, system design, and implementation strategies presented in this study served as a valuable guide for researchers who undertook similar projects in the future. Additionally, future researchers could use this study to identify the limitations and challenges encountered during development, which helped them formulate better approaches and more effective solutions. The study also opened opportunities for further exploration of emerging technologies such as artificial intelligence, machine learning, and cloud-based systems that could be integrated into property management platforms. Moreover, the data and insights gathered from this study contributed to the growing body of knowledge in the field of information systems and web-based business solutions, supporting the advancement of future academic and professional research.

• **Academic Institution** – The institution benefited from this study as an additional academic resource that demonstrated the practical use of information systems in solving business operational challenges and promoting innovation among students. Furthermore, the study served as a testament to the institution's commitment to producing competent and skilled graduates who were capable of developing practical technology solutions for real-world business needs. The completed system and its documentation were made available as reference materials that future students and faculty members could utilize for academic purposes and further research. Additionally, the institution was able to showcase this study as an example of student achievement that reflected the quality and effectiveness of its information technology curriculum. The study also contributed to the institution's academic resources by providing a well-documented case of applied systems development, which supported instructors in teaching relevant concepts in web

development, database management, and systems analysis. Moreover, the success of this study encouraged more students to engage in research and development projects, fostering a culture of innovation and academic excellence that strengthened the institution's reputation in the field of information technology.

1.6 Definition of Terms

The following terms are defined operationally, within the context of the Condominium Booking and Operations Management System developed for Gales Urban Haven (One Spatial Iloilo), to guide the reader's interpretation of this paper.

- **Automation**: the use of software logic to execute repetitive, rule-based operational processes with minimal manual intervention. In this study, automation is applied to booking processing, payment computation and recording, financial report generation, and maintenance scheduling. It reduces staff workload, minimizes human error, and lets staff spend more time on tasks that require judgment.
- **Booking Management System**: the module that records, organizes, processes, and monitors reservation transactions, including booking details, tenant information, unit availability, and reservation history. It replaces the manual booking process of the business. It offers search and filtering by period, unit type, or tenant profile, and it connects with the payment and tenant records to give administrators a complete view of ongoing and completed reservations.
- **Broadcast Console**: the communication interface through which authorized administrators compose, schedule, and send announcements, notices, billing advisories, and maintenance alerts to individual units, groups of units, or all residents. Each message is logged in the audit trail and its delivery status is tracked. It replaces informal channels such as verbal relays and untracked messaging applications.
- **Calendar Integration**: the scheduling module that presents reservations, check-ins and check-outs, unit availability windows, and maintenance activities in a single interactive calendar. Users can filter events by date, unit, or event type. The calendar helps prevent double bookings, generates reminders for upcoming events, and gives staff a shared, current view of operations.
- **Cash Ledger**: the financial record within the system that documents all monetary transactions per tenant and unit, including down payments, installment payments, penalty charges, adjustments, and outstanding balances. It posts each transaction as it occurs, computes running balances, flags overdue accounts, and generates account statements on demand. It is the primary data source for revenue monitoring.

- **Centralized Database**: the single, unified repository that stores booking records, tenant profiles, payment transactions, financial summaries, and maintenance logs. All modules read from and write to it, which eliminates data silos and redundancy. It applies access controls and serves as the long-term archive for reference, auditing, and analysis.
- **Condominium Booking and Operations Management System**: the web-based information system developed in this study to integrate booking management, payment monitoring, revenue tracking, calendar-based scheduling, and maintenance management in one platform. It is designed for the operational needs of Gales Urban Haven and gives owners and staff real-time access to operational data without cross-referencing multiple records.
- **Data Accuracy**: the degree to which recorded information is correct, consistent, and free of errors. In this study it covers booking details, payment figures, tenant records, and maintenance logs. The system supports it through input validation, automated computation, structured data entry, duplicate detection, and audit trails of changes to critical records.
- **Efficiency**: the capacity to execute business processes accurately and quickly with minimal wasted resources. In this study it refers to the improvement in the speed, accuracy, and reliability of the operations of Gales Urban Haven attributed to the proposed system.
- **Maintenance Tracking**: the module for logging, scheduling, monitoring, and updating maintenance requests, repairs, preventive maintenance schedules, and service completion status. Administrators can assign tasks, set priority levels, and maintain a complete maintenance history. The module also produces reports on recurring issues and pending requests.
- **Payment Tracking**: the module that monitors and records tenant payments, including down payments, installments, outstanding balances, and payment histories. It flags overdue accounts, computes penalty fees, and produces payment status reports and tenant-specific summaries. It also links to the booking and tenant records so that financial and operational data remain consistent.
- **Prototype**: the initial working version of the system, presented to end users and evaluators for testing, demonstration, and feedback before full deployment. It allows design flaws, functional gaps, and usability issues to be found early, when they are cheaper to correct.
- **Resident Directory**: the registry within the system of all registered residents and active tenants of Gales Urban Haven. Each profile holds biographical and contact information, unit assignment, tenancy classification, move-in and move-out dates, verification status,

and linked financial accounts. Other modules, including the cash ledger, broadcast console, and audit trail, rely on it to attribute records to the correct person and unit.

- **Revenue Monitoring**: the automated process of recording, aggregating, and computing income from all booking transactions, including payments received, down payments collected, and balances pending. It helps prevent income leakage and undetected discrepancies, and it supports revenue reports and trend analysis for decision-making.
- **System Audit Trail**: the chronological electronic record of user actions across all modules, including logins and logouts, booking changes, ledger postings, broadcast messages, and directory updates. Each entry records the user, the action, the affected data fields, and the timestamp. Access is restricted to users with audit or administrative privileges. The trail supports accountability, discrepancy resolution, and reconstruction of the history of any transaction.
- **User Authentication**: the access control mechanism that verifies a user's identity through valid credentials before granting access to system features and data. It supports role-based access control so that each user sees only the features relevant to their responsibilities. The system also keeps an access log of successful and failed login attempts.
- **Web-Based System**: a software application hosted on a web server and accessed through a standard browser without installing dedicated software on the user's device. It allows multiple authorized users to work simultaneously from different locations, keeps data centralized and synchronized, and simplifies updates and maintenance.

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CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 Foreign Studies

Imkumnert [1] proposed a system that handles most of the information management in a condominium, such as customer, room, and service information, room renting, billing and receipt operations, and reports for the management team. Rental condominiums are becoming popular in the lodging market. The real-estate business had previously focused on the condominium market through a construction and selling methodology.

Halkiopoulos et al. [2] explored knowledge of online booking systems and the views of local student-users on booking rates through these systems. They also investigated the emotion-focused decision-making process that users follow when choosing a tourist destination through online booking systems.

Varutbangkul [3], at Assumption University, implemented a condominium management information system to improve the efficiency of the existing system at Fair Tower, a condominium belonging to GOS Limited Partnership. The project designed a new computerized system to replace the old manual system at the front office, so that front-desk staff could work more efficiently and accurately. The proposed system handles most of the information management in the condominium, including customer, room, and service information, room sales, room renting, bills and receipts, and reports for the general manager.

Kuo and Chou [4] studied condominium management (CM) businesses, which rely on continuous improvement of service quality (SQ) to increase consumer satisfaction (CS) and thereby improve operations. A lack of knowledge about SQ and CS results in poor efficiency, so the relationship between the two must be investigated to develop cost-effective improvement strategies. They first used exploratory factor analysis to cluster similar attributes into three constructs and identify significant attributes for assessment. Resident satisfaction scores (RSSs) were then computed to assess the relative importance rate (RIR) of the attributes, using an adapted multivariate method that regresses the RSSs on SQ attribute performance. Finally, a service-quality model integrating an improved Kano's approach with importance–performance

analysis (IPA) was used to develop a prioritized strategy for improving satisfaction in the CM business.

2.2 Local Studies

Bajado [5] investigated the impact of hotel business tools and technologies on guest satisfaction in Tagaytay City, Philippines, a rapidly growing tourist destination. Hotels worldwide are adopting technology to enhance guest experiences and satisfaction.

Gosela and Encarnacion [6] presented a unified application that addresses existing challenges through user-centric features. Using location-based services and streamlined booking processes, the application aims to improve efficiency, accessibility, and sustainability in the tourism sector. It provides real-time tracking, efficient transit solutions, and a comprehensive directory of establishments.

Gallenero et al. [7] developed a real-estate management system with sales analysis and customer segmentation. Rising population increases the demand for real estate, and managing real estate manually takes considerable time and manpower. Manual handling is also prone to damage and misplacement of documents and client data. The system automates complex tasks and duties and also considers customer support and community. Customer segmentation was used for sales analysis to improve sales and to understand the company's current, possible, and future customers. The system was developed using the Agile methodology, with users holding different roles and administrator rights to access and manage it. Testing, verification, validation, usability testing, and acceptance testing, including alpha and beta testing, were conducted to ensure the requirements were met.

2.3 Related Literature

This chapter presents a comprehensive review of related literature that serves as the theoretical and empirical foundation for the development of the Condominium Booking and Operations Management System proposed in this study. The literature reviewed covers a broad range of thematic areas directly relevant to the proposed system: property management systems, automated booking and reservation technologies, payment tracking and revenue monitoring, maintenance management, centralized database systems, web-based application development, and the application of the Agile development methodology in information systems projects. The insights and findings drawn from the reviewed literature collectively inform the design decisions, functional specifications, and architectural choices that underpin the proposed system for Gales Urban Haven, operating under the business name One Spatial Iloilo.

Property Management Systems

Property management systems (PMS) have been the subject of extensive scholarly and professional inquiry over the past two decades, owing to the growing recognition that the efficient management of residential, commercial, and mixed-use properties requires the support of robust, integrated digital tools. The literature broadly defines a property management system as a software platform that consolidates the core operational functions of a property management business into a single, centralized, and accessible interface. These functions include tenant record management, lease administration, financial tracking, maintenance scheduling, and reporting.

Bardi [8] affirmed that property management systems have fundamentally transformed the hospitality and real estate industries by replacing fragmented, paper-based administrative processes with automated, data-driven workflows. These workflows significantly reduce operational errors and improve service delivery standards. The author noted that integrating booking, billing, and reporting functions within a unified platform enables property managers to maintain a comprehensive and current view of all operational activities, thereby improving their capacity for informed decision-making and proactive problem resolution.

Tesone [9] similarly argued that modern property management systems represent an indispensable operational infrastructure for any property management enterprise operating in a competitive market environment. His research emphasized that the ability to retrieve accurate and up-to-date information across all operational domains, from reservation status to financial performance, at any time and from a single platform, is a decisive competitive advantage over organizations that continue to rely on manual methods.

In the Philippine context, the Department of Information and Communications Technology (DICT) [10] has acknowledged the growing importance of information and communications technology adoption among small and medium enterprises (SMEs) in the service sector. It noted that businesses in property management, hospitality, and rental services that adopt digital management systems report significantly higher levels of operational efficiency, financial accuracy, and customer satisfaction than their non-digitalized counterparts. This observation is particularly relevant to the present study, given that Gales Urban Haven is a small-scale condominium management operation whose administrative processes remain largely manual and are therefore vulnerable to the inefficiencies that digital systems are specifically designed to address.

Cobanoglu et al. [11] further highlighted that the effectiveness of a property management system is determined not solely by its technological sophistication but also by how well its design and functionality align with the specific operational workflows and data management needs of its target users. This finding underscores the importance of a thorough requirements analysis prior to system development and of involving end users in the design and evaluation process. Both are key activities in the Agile development approach adopted in the present study.

Automated Booking and Reservation Management

The automation of booking and reservation processes has been extensively studied in both the hospitality and real estate management literature. A consistent body of evidence shows that automated reservation systems substantially outperform manual methods in speed, accuracy, reliability, and scalability. The literature identifies scheduling conflicts, double bookings, and incomplete reservation records as the most common and consequential problems of manual booking systems, and affirms that automated booking management platforms effectively mitigate them.

O'Connor and Murphy [12] studied the adoption of online reservation systems in the hospitality sector. They found that properties using automated booking platforms reported a significant reduction in reservation errors, faster booking processing, and a marked improvement in the accuracy of unit availability information available to both administrators and clients. The authors concluded that real-time availability checking is among the most operationally impactful features of such systems, as it eliminates the information lag inherent in manual availability verification.

Siguaw and Enz [13] similarly documented that implementing computerized reservation systems in lodging establishments produced measurable improvements in front-desk efficiency, with staff able to process reservations in a fraction of the time previously required. Their findings further indicated that automated reservation systems gave managers easier access to historical booking data, facilitating trend analysis and demand forecasting that were previously impractical under manual record-keeping.

Kasavana and Brooks [14] noted that the fundamental advantage of an automated booking system lies in its capacity to maintain a continuously updated, centrally accessible, and conflict-free reservation schedule. The system generates an alert or blocks the entry if two reservations are recorded for the same unit at the same time. This conflict-prevention mechanism, a standard feature of well-designed booking management systems, directly addresses one of the most critical risks of manual reservation processes: the inadvertent recording of overlapping bookings that results in tenant disputes and administrative complications.

For small-scale condominium and apartment rental management, Pizam and Ellis [15] emphasized that the benefits of booking automation are proportionally more significant for smaller properties than for large hotel chains. Smaller operations typically have fewer administrative staff and therefore less capacity to absorb the time and error costs of manual booking management. This observation applies directly to Gales Urban Haven, where the administrative team is small and the demands of managing multiple condominium units simultaneously require a system that can handle reservation data with speed, accuracy, and minimal staff intervention.

Payment Tracking and Financial Record Management

The management of tenant payments, including the recording of initial deposits, monitoring of outstanding balances, and maintenance of complete payment histories, is one of the most critical and error-sensitive functions in any property rental operation. The literature on financial record management in property contexts consistently identifies the absence of a systematic, automated payment tracking mechanism as a primary source of financial discrepancies, revenue leakage, and administrative disputes between property managers and tenants.

Warren and Jones [16] studied financial management practices among small residential rental operations. They found that businesses relying on manual payment recording experienced significantly higher rates of recording errors, missed payment notifications, and balance computation discrepancies than those using automated financial tracking tools. The authors concluded that even relatively simple automated payment tracking systems substantially improve financial accuracy and accountability, reduce the administrative burden of manual reconciliation, and improve the reliability of financial records available to management.

Clauretie and Sirmans [17] further argued that accurate and timely payment tracking is not merely an administrative convenience but a strategic financial management imperative for property rental businesses. The accuracy of payment records directly determines the reliability of cash flow projections, revenue reports, and business planning decisions. Their research showed that property managers who maintained comprehensive, up-to-date payment records were significantly better positioned to make informed decisions on pricing, capacity expansion, and operational investment than those whose financial records were incomplete or delayed.

Turner [18] examined the integration of payment tracking with broader financial management systems in property management operations. He found that consolidating payment data, revenue figures, and financial reporting within a single platform eliminated the data reconciliation delays that commonly arise when these functions are managed through separate, disconnected tools. This finding is directly relevant to the proposed system for Gales Urban Haven, in which the payment tracking module will be fully integrated with the revenue reporting module so that all financial data is processed, stored, and reported within a single, cohesive system environment.

Revenue Monitoring and Financial Reporting

Revenue monitoring and financial reporting are foundational functions of any business operation, and their importance in property management contexts has received considerable scholarly attention. The literature consistently emphasizes that accurate, timely, and accessible financial reporting is essential for enabling property management organizations to assess their financial performance, identify revenue trends, manage cash flow effectively, and make evidence-based strategic decisions.

Brigham and Houston [19] described revenue monitoring as the systematic and continuous process of tracking all income streams, computing aggregate revenue across defined time periods,

and comparing actual revenue performance against projected targets. Their work emphasized that automated revenue monitoring systems, which perform these functions without manual data compilation, give management access to accurate financial performance data in real time. This dramatically reduces the lag between financial activity and financial awareness that characterizes manual reporting.

Horngren et al. [20] further noted that the quality of management decision-making is directly and substantially influenced by the timeliness and accuracy of the financial information available to decision-makers. Their research showed that organizations with real-time financial reporting tools consistently outperformed those relying on periodic manual reports in identifying and responding to financial challenges before they escalate into serious operational problems. This finding strongly supports the inclusion of an automated revenue reporting module in the proposed system for Gales Urban Haven, where the current absence of such a tool has been identified as a significant impediment to effective financial management.

Kimmel et al. [21] similarly highlighted the importance of financial report accuracy in small business contexts. They noted that errors in manually prepared financial reports can have disproportionately serious consequences for small enterprises, given their typically limited financial reserves and narrower margins for error. The authors advocated the adoption of automated financial reporting tools by small business operators as a critical risk management strategy. This recommendation directly reinforces the rationale for incorporating revenue reporting automation into the proposed system for One Spatial Iloilo.

Maintenance Management in Property Operations

Maintenance management is widely recognized in the property management literature as a critical determinant of property value retention, tenant satisfaction, and operational continuity. The systematic recording, scheduling, monitoring, and resolution of maintenance activities is essential for keeping condominium units and shared facilities in optimal condition throughout their operational lifecycle.

Dhillon [22] provided a comprehensive framework for maintenance management in built environments, arguing that the effectiveness of a maintenance program is directly determined by the quality of the systems and processes used to track and manage maintenance activities. His research showed that organizations using structured, digitally supported maintenance management systems consistently achieved higher rates of timely maintenance resolution, fewer repeated maintenance failures, and higher tenant satisfaction than those relying on informal or paper-based tracking.

Mobley [23] similarly emphasized the strategic importance of preventive maintenance scheduling in property management. He noted that a systematic approach to maintenance planning, supported by accurate historical maintenance records and a reliable scheduling system, enables property managers to address potential issues before they develop into costly emergency

repairs. His work highlighted the role of digital maintenance logs in helping managers identify patterns of recurring problems and make data-driven decisions about maintenance resource allocation and preventive intervention priorities.

Wireman [24] further documented that the absence of a systematic maintenance tracking system in property management operations frequently leads to the loss of maintenance history records. This makes it difficult for managers to assess the long-term performance of maintenance interventions, track the lifecycle of property assets, and demonstrate maintenance accountability to property owners. These findings provide strong empirical support for a dedicated maintenance monitoring module within the proposed system for Gales Urban Haven, ensuring that all maintenance activities are systematically recorded, tracked, and accessible to administrators for review at any time.

Calendar-Based Scheduling in Operations Management

Calendar-based scheduling tools have been widely studied in operations management across multiple industries, including hospitality, healthcare, education, and property management. The literature consistently affirms that visual scheduling interfaces give administrators a significant operational advantage by consolidating time-sensitive operational data into a single, chronologically organized display that enables rapid situational awareness and more effective operational planning.

Gaither and Frazier [25] noted that scheduling is among the most cognitively demanding and error-prone functions in operations management when performed manually. It requires administrators to simultaneously track and reconcile multiple overlapping activities, availability windows, and time constraints without the support of a visual organizational framework. Their research showed that digital scheduling tools reduce the cognitive load of scheduling tasks and substantially decrease the frequency of scheduling errors and conflicts in complex operational environments.

Pinedo [26] further documented that integrated scheduling systems, which synchronize calendar data with other operational modules such as reservations, availability tracking, and maintenance records, provide the most significant operational benefits compared to standalone scheduling tools. Synchronizing scheduling data with real-time operational information ensures that the calendar view presented to administrators always reflects the current state of operations, enabling scheduling decisions based on reliable, up-to-date information. This principle directly informed the design of the calendar scheduling module in the proposed system, which will be fully integrated with the booking and maintenance modules to ensure real-time data consistency across all scheduling displays.

Centralized Database Systems in Information Management

The design and implementation of centralized database systems has been a central area of inquiry in information systems research. The literature consistently affirms that centralized relational

databases provide a superior data management foundation for multi-functional information systems compared to fragmented or distributed data storage approaches. Centralized databases enable organizations to eliminate data redundancy, enforce data consistency and integrity, and give all authorized users access to the same, current version of organizational data regardless of their location or device.

Date [27] provided a seminal treatment of relational database theory, arguing that the relational model, implemented through systems such as MySQL, provides the most robust and flexible framework for organizing complex, interrelated data sets in organizational contexts. His work emphasized that structured query language (SQL) enables powerful and precise data retrieval that would be impractical or impossible in non-relational environments, including the spreadsheet-based and paper-based systems commonly used by small property management operations.

Elmasri and Navathe [28] further documented the importance of database normalization in ensuring the integrity and consistency of data in relational databases. They noted that a properly normalized schema eliminates data redundancy, prevents update anomalies, and keeps all data within the system internally consistent. Their work also highlighted the importance of appropriate indexing strategies to optimize query performance and keep the system responsive as data volume grows. This is particularly relevant to the proposed system for Gales Urban Haven, which will accumulate increasingly large volumes of booking, payment, and maintenance records over its operational life.

Silberschatz et al. [29] addressed the security dimensions of centralized database management. They emphasized that robust access control mechanisms, parameterized query execution, and regular data backup protocols are essential components of any database system that handles sensitive organizational or financial data. Their recommendations directly inform the security specifications in the database design of the proposed system, which will implement role-based access controls, prepared statement query mechanisms, and scheduled backup procedures to protect all operational data stored for One Spatial Iloilo from unauthorized access, data corruption, and accidental loss.

Web-Based Information Systems

Web-based information systems have become the dominant architectural paradigm for organizational information management across virtually all industry sectors, owing to their platform independence, accessibility, scalability, and ease of maintenance. The literature consistently documents their advantages over locally installed desktop applications, particularly where multiple users need to access and interact with shared data simultaneously from different workstations or locations.

Laudon and Laudon [30] provided a comprehensive treatment of management information systems in organizational contexts, arguing that web-based systems are the most effective and

adaptable information management infrastructure available to modern organizations. Their work highlighted that web-based systems let organizations deploy and update their information systems more efficiently than locally installed alternatives. Modifications need only be applied at the server level and are immediately available to all users without software updates on individual client devices.

O'Brien and Marakas [31] similarly emphasized that web-based systems offer a critical advantage in data consistency. All users access the same centrally hosted application and data repository, which eliminates the synchronization problems that arise when multiple users maintain separate local copies of organizational data. This advantage is particularly significant in property management operations, where multiple staff members may need to access and update shared booking, payment, and maintenance records simultaneously, and where inconsistencies from unsynchronized local copies can lead to serious operational errors.

Turban et al. [32] further noted that the accessibility of web-based systems through standard browsers, without specialized software installation, significantly lowers the barrier to adoption for small and medium enterprises. It eliminates the hardware and software procurement costs commonly associated with proprietary desktop deployments. This is particularly relevant to One Spatial Iloilo, where a web-based system will enable the organization to implement a comprehensive digital management platform without the substantial upfront costs of traditional enterprise software installations.

Agile Software Development Methodology

The Agile software development methodology has been extensively studied and widely adopted in the information systems development community since the publication of the Agile Manifesto in 2001. The Manifesto articulated a set of values and principles prioritizing iterative development, customer collaboration, responsiveness to change, and the continuous delivery of working software over the rigid adherence to comprehensive upfront planning that characterizes traditional development models.

Pressman and Maxim [33] provided a thorough examination of Agile development practices in their software engineering textbook. They documented that Agile methodologies consistently produce higher-quality software than traditional Waterfall-based approaches when applied to projects with evolving or initially unclear requirements. Their analysis showed that the iterative structure of Agile development enables teams to incorporate user feedback and stakeholder input at each stage, resulting in systems that more closely match the actual needs and expectations of their intended users at delivery.

Sommerville [34] similarly noted that Agile development is particularly well suited to projects in which the full scope of system requirements cannot be determined at the outset, as its iterative cycles allow requirements to be progressively clarified, refined, and expanded as development proceeds and users better understand the system's capabilities and their own operational needs.

This makes Agile especially appropriate for the proposed system for Gales Urban Haven. The complexity of integrating multiple operational modules, and the need to align system functionality with the nuanced and evolving workflows of the property management team, make iterative development and continuous feedback essential to the success of the project.

Schwaber and Sutherland [35], the co-creators of the Scrum framework, one of the most widely used implementations of the Agile methodology, documented that organizations applying Agile practices in information systems development report higher rates of project completion within budget and schedule, higher end-user satisfaction with the delivered system, and greater capacity to adapt to changing requirements than organizations using traditional methodologies. These documented outcomes provide strong empirical support for selecting the Agile model as the guiding development methodology for the proposed Condominium Booking and Operations Management System.

2.4 Synthesis of the Reviewed Literature

The body of literature reviewed in the preceding sections of this chapter collectively established a comprehensive and well-grounded theoretical and empirical foundation for the development of the Condominium Booking and Operations Management System proposed in this study. Across the various domains examined, including property management systems, booking and reservation technologies, payment and revenue tracking, maintenance management, database architecture, web-based system design, and the application of the Agile development methodology, a consistent and converging set of insights emerged. These insights affirmed that the operational challenges experienced by Gales Urban Haven under One Spatial Iloilo were not unique to the organization. They were, rather, characteristic of a broader class of problems that had been extensively documented in the academic and professional literature on small- to medium-scale property management operations that continued to rely on manual or fragmented record-keeping systems.

The synthesis of these reviewed works revealed that the transition from manual, paper-based, or loosely organized digital processes to a centralized, automated, and integrated information system was not merely a technological upgrade but a fundamental transformation in the way a property management business operated, managed its resources, served its tenants, and positioned itself for sustainable growth. The literature consistently demonstrated that organizations that had undertaken this transition realized measurable improvements in operational efficiency, financial accuracy, data accessibility, staff productivity, and overall service quality. The proposed system for Gales Urban Haven was firmly grounded in these documented outcomes, and its design, features, and development approach were shaped directly by the collective knowledge and evidence presented in the reviewed literature.

Convergence of Key Themes Across the Literature

A careful examination of the reviewed literature revealed that despite the diversity of contexts, geographic settings, and technological environments in which the individual studies were conducted, a set of dominant and recurrent themes emerged with striking consistency. These themes not only validated the problem statement articulated in Chapter One of this study but also provided direct and actionable guidance for the scope, design, and functionality of the proposed system. The following subsections discussed these converging themes in detail and explained how each one informed the development of the Condominium Booking and Operations Management System.

Furthermore, the convergence of these themes across multiple studies and literature sources strengthened the theoretical and empirical foundation of the proposed system, demonstrating that the identified operational challenges faced by Gales Urban Haven were not isolated concerns but widely recognized issues that had been consistently addressed through the adoption of web-based information systems. The recurring patterns observed across the reviewed literature also served as a guiding framework in determining the most essential features and functionalities that the proposed system needed to incorporate in order to respond effectively to the operational demands of a condominium rental business.

Additionally, the synthesis of these converging themes highlighted the critical role that automation, centralized data management, and real-time monitoring played in modernizing property management operations, further justifying the relevance and necessity of the proposed system. The alignment between the findings of the reviewed literature and the operational context of Gales Urban Haven also reinforced the appropriateness of the chosen system design and development approach, ensuring that the proposed solution was grounded in established knowledge and proven practices. Moreover, the identification of these key themes allowed the researchers to draw meaningful connections between existing studies and the specific objectives of this research, providing a coherent and well-supported academic basis for the development and implementation of the Condominium Booking and Operations Management System.

Ultimately, the convergence of key themes across the literature affirmed that the proposed system represented a timely, relevant, and well-informed response to the operational inefficiencies of Gales Urban Haven. It contributes not only to the improvement of its business processes but also to the broader academic discourse on the application of information systems in property management and hospitality operations.

The Pervasive Inadequacy of Manual Property Management Processes

Perhaps the most consistently documented finding across the reviewed literature was the fundamental inadequacy of manual, paper-based, and non-integrated approaches to property management. Studies examining condominium associations, residential rental properties, commercial real estate portfolios, and mixed-use developments all converged on the conclusion

that organizations relying on such approaches routinely suffered from a predictable constellation of operational failures. These included delayed and error-prone tenant billing, inconsistent maintenance request tracking, inability to generate accurate real-time financial reports, poor visibility into unit occupancy and availability, and fragmented communication between property managers, maintenance staff, and tenants.

The literature made clear that these failures were not isolated incidents but the natural and expected consequences of systems designed for lower volumes of transactions and simpler operational environments. As the scale and complexity of property portfolios grew, in terms of the number of units managed, the diversity of tenant needs, the variety of services offered, and the stringency of regulatory compliance requirements, the limitations of manual systems became increasingly acute. For Gales Urban Haven, which operated in a growing urban condominium market in Iloilo City, these dynamics were particularly relevant and underscored the urgency and necessity of the proposed system.

The Demonstrated Value of Centralized and Integrated Information Systems

In direct response to the documented inadequacies of manual approaches, the literature overwhelmingly endorsed the adoption of centralized, integrated property management information systems as the most effective and sustainable solution. The reviewed studies documented concrete and quantifiable benefits realized by organizations that had implemented such systems, including reductions in administrative workload, improvements in billing accuracy and collection rates, faster resolution of maintenance issues, and enhanced tenant satisfaction scores.

Centralization was identified as a particularly critical success factor. When all operational data, including tenant records, unit availability, booking schedules, payment histories, maintenance requests, and financial summaries, resided in a single, authoritative system of record, the risks of data duplication, inconsistency, and loss were dramatically reduced. Property managers were able to access up-to-date information at any time, from any location, enabling more responsive and informed decision-making. The proposed Condominium Booking and Operations Management System embodied this principle of centralization as one of its foundational design objectives.

The Strategic Importance of Booking and Reservation Management

The literature reviewed on booking and reservation systems established that the management of unit availability and the processing of booking transactions represented some of the most operationally critical and error-prone functions in property management. Scheduling conflicts, double bookings, and the failure to maintain accurate availability calendars were identified as frequent sources of tenant dissatisfaction, revenue loss, and reputational damage for property management organizations.

The reviewed studies demonstrated that automated booking systems equipped with real-time availability tracking, conflict detection, and instant confirmation capabilities effectively eliminated or significantly reduced these problems. The literature further established that integrating booking management with tenant records and payment processing created a seamless operational workflow that reduced the manual effort required to complete a booking transaction from end to end. These documented outcomes directly informed the booking management module that was developed as a core component of the proposed system for Gales Urban Haven.

The Critical Role of Accurate Financial and Payment Tracking

Across the financial management domain, the reviewed literature identified inaccurate and incomplete payment tracking as one of the most consequential operational deficiencies in property management organizations. The inability to accurately record, reconcile, and report on rent collection, association dues, security deposits, utility charges, and penalty fees created compounding problems that affected cash flow management, financial planning, and regulatory compliance. The literature documented that automated payment tracking systems, particularly those that generated digital receipts, sent payment reminders, flagged overdue accounts, and produced summary financial reports, consistently outperformed manual equivalents in collection rates, accuracy, and audit readiness. The proposed system incorporated a comprehensive payment and revenue tracking module informed by these findings, ensuring that all financial transactions conducted through Gales Urban Haven were recorded with precision, transparency, and full auditability.

As the system became fully operational, the payment tracking module served as a critical instrument in strengthening the financial governance and accountability practices of the organization, providing administrators with a reliable and centralized platform for monitoring all incoming and outgoing financial transactions in real time. The structured financial data generated by the system also enabled management to produce accurate periodic financial statements, facilitating more informed budgeting decisions and long-term financial planning that supported the sustainable growth of Gales Urban Haven. The payment tracking module also incorporated advanced analytics capabilities that allowed administrators to identify payment trends, forecast revenue streams, and detect anomalies or irregularities in financial records before they escalated into more serious concerns.

The system further supported the generation of tenant-specific payment histories, which served as valuable documentation in cases of financial disputes, lease renewals, or regulatory audits conducted by relevant government authorities. Integrating the payment tracking module with other system components, such as booking management and tenant records, created a fully interconnected financial ecosystem that eliminated data redundancy and ensured consistency across all financial and operational records maintained by the organization. The adoption of a digitized and automated payment tracking approach also significantly reduced the administrative burden placed on staff, freeing time and resources that were redirected toward higher-value

management activities and tenant relationship building. Ultimately, the financial and payment tracking capabilities embedded within the system laid a strong and enduring foundation for the continued improvement of fiscal responsibility, operational transparency, and financial performance at Gales Urban Haven.

Maintenance Management as a Driver of Tenant Retention

The reviewed literature on maintenance management revealed a strong and statistically significant relationship between the responsiveness of a property's maintenance operations and the satisfaction and retention rates of its tenants. Properties that implemented structured, technology-supported maintenance request workflows, including formal request submission channels, automated routing to appropriate staff, status tracking, and completion confirmation, were shown to resolve issues faster, reduce the number of repeat complaints for the same issue, and generate higher tenant satisfaction ratings than properties relying on informal or verbal maintenance reporting.

These findings had direct implications for the design of the maintenance management module included in the proposed system. The module was designed to provide tenants with a formal and accessible channel for submitting maintenance requests and tracking their progress, while providing property managers and maintenance personnel with a structured workflow for prioritizing, assigning, and documenting the resolution of reported issues.

Database Architecture as the Backbone of System Reliability

The literature reviewed on database systems and architecture established that the choice of database technology and schema design had profound implications for the reliability, performance, scalability, and data integrity of any information system. The reviewed studies consistently recommended relational database management systems (RDBMS) for applications that managed structured, interdependent data entities with complex relationships, a description that accurately characterized the data model of a property management system.

Key principles drawn from the database literature were all applied in the design of the database layer of the proposed system. These include normalization to eliminate data redundancy, enforcement of referential integrity through foreign key constraints, indexing to optimize query performance, and transaction management to prevent data corruption. These design decisions ensured that the system maintained the integrity and consistency of all stored data even under concurrent access from multiple users.

Web-Based Accessibility as an Enabler of Operational Continuity

The reviewed literature on web-based information systems established that the accessibility and platform independence afforded by browser-based applications represented a significant operational advantage over desktop-installed or server-bound legacy systems. Web-based systems allowed authorized users to access the system from any device with an internet

connection, eliminating the need for dedicated hardware installations, simplifying software updates and maintenance, and enabling remote management of property operations.

For a property management context, this accessibility was particularly valuable. Property managers were able to respond to booking inquiries, review payment statuses, and monitor maintenance requests from outside the office. Tenants were able to submit requests, view their payment history, and communicate with management without having to visit a physical office. These capabilities directly improved the operational responsiveness and service quality of Gales Urban Haven, and the proposed system was designed as a web-based application to realize these benefits fully.

Agile Methodology as the Appropriate Development Framework

The literature reviewed on software development methodologies provided strong justification for selecting the Agile methodology as the development framework for the proposed system. The Agile approach, characterized by iterative development cycles, continuous stakeholder collaboration, adaptive planning, and incremental delivery of working software, was shown to be particularly well suited for projects in which the full scope of requirements could not be exhaustively defined at the outset, a condition that accurately described the development context of the proposed system.

The reviewed studies documented that Agile projects consistently achieved higher rates of stakeholder satisfaction, more timely delivery, and better alignment between the delivered system and actual user needs compared to projects managed under the traditional Waterfall methodology. By adopting Agile, the development team was positioned to respond constructively to feedback from the management and staff of Gales Urban Haven throughout the development process, ensuring that the final system accurately reflected the operational realities and priorities of the organization.

Research Gaps Addressed by the Proposed System

While the reviewed literature provided robust evidence for the value of property management information systems in general, it also revealed a number of research and implementation gaps that the proposed system was positioned to address. These gaps related to the specific context of small- to medium-scale condominium operations in emerging urban markets in the Philippines, the integration of multiple management functions within a single unified system, and the accessibility of such systems to organizations with limited technical infrastructure.

Theoretical Frameworks Informing the Proposed System

Beyond the empirical findings reported in individual studies, the reviewed literature also surfaced several theoretical frameworks and conceptual models that provided a principled basis for the design decisions incorporated in the proposed system. These frameworks guided not only what the system did but also how it was structured, how its components related to one another,

and how it was evaluated upon completion. The theoretical foundations ensured that the proposed system was not merely a technical solution to immediate operational problems but a thoughtfully designed platform grounded in established principles of information systems development, property management theory, and organizational efficiency. Each framework contributed a distinct perspective that informed specific aspects of the system architecture, user interface design, data management approach, and overall functional scope.

The application of these conceptual models also provided a systematic basis for justifying the design choices made throughout the development process, ensuring that every major decision was supported by both theoretical reasoning and empirical evidence drawn from existing literature. Furthermore, integrating multiple theoretical frameworks allowed the development team to approach the system design from a multidimensional perspective, accounting for the technical, organizational, and human factors that collectively determined the effectiveness and sustainability of the proposed system in a real-world property management environment.

As the system progressed through its development and evaluation phases, these frameworks continued to serve as reference points against which the system's performance, usability, and alignment with organizational goals were measured. Future researchers and developers who built upon the findings of this study could also benefit from the theoretical groundwork established herein, as it provided a coherent and well-documented intellectual foundation for subsequent enhancements, adaptations, and expansions of the proposed system. Ultimately, the deliberate incorporation of theoretical frameworks into the system design process elevated the academic rigor and practical relevance of this capstone project, demonstrating that the proposed solution was not only technically functional but also conceptually sound and aligned with the broader body of knowledge in information technology and property management.

Systems Theory and the Integrated Operations Model

The concept of the organization as an integrated system, in which all operational processes were interdependent and the performance of the whole was a function of the coherence of its parts, provided the theoretical foundation for the integrated design of the proposed system. Rather than developing separate standalone tools for booking, payment, and maintenance, the proposed system was designed as an interconnected platform in which data generated by one module was immediately available to and usable by all other modules. This integrative approach, grounded in systems theory, eliminated the data silos that were identified as a primary source of inefficiency in the reviewed literature.

The application of systems theory ensured that each module operated not as an isolated component but as a contributing element of a larger, cohesive operational framework that served the holistic information and management needs of Gales Urban Haven. By treating the organization as a unified system of interdependent processes, the proposed system was capable

of reflecting the true complexity of condominium property management operations, where a single transaction or event in one area of the business inevitably produced ripple effects across multiple other functional domains. The interconnected architecture also enabled the system to maintain a single, consistent, and continuously updated data environment, ensuring that all modules operated on accurate and current information rather than fragmented or duplicated records maintained across separate platforms.

As the system was adopted and utilized, the systemic coherence built into its design contributed to measurable improvements in the speed, accuracy, and reliability of administrative decision-making, as administrators were able to access a comprehensive and integrated view of all operational activities from a single centralized interface. The systems theory foundation also provided a scalable and flexible structural framework that accommodated the addition of new modules and functionalities without compromising the integrity or coherence of the overall platform. It also informed the evaluation criteria used to assess the success of the proposed system, with particular emphasis on the degree to which integration across modules reduced redundancy, improved data consistency, and supported more efficient and informed organizational management. Ultimately, grounding the system in systems theory strengthened its technical design and kept it aligned with the operational realities and strategic goals of Gales Urban Haven.

The Technology Acceptance Model (TAM)

The Technology Acceptance Model, which posited that the adoption and effective use of an information system were determined primarily by users' perceptions of its usefulness and ease of use, informed the user experience design strategy of the proposed system. In line with the TAM framework, the system was designed with a strong emphasis on interface clarity, logical workflow navigation, and minimal cognitive load for end users. The user interface was iteratively refined through stakeholder feedback sessions conducted during the Agile development process, ensuring that the delivered system met the usability standards necessary for adoption by all categories of users at Gales Urban Haven. Furthermore, the principles of the Technology Acceptance Model guided the researchers in evaluating the degree to which the proposed system was perceived as both useful and easy to use by its intended users, providing a structured basis for assessing user satisfaction and system acceptance upon implementation.

The TAM framework also informed the design of system features that directly addressed the practical needs of administrators and business owners, ensuring that every functional module contributed meaningful value to their daily operational tasks and decision-making processes. Additionally, the emphasis on perceived ease of use drove the development of an intuitive and straightforward user interface that minimized the time and effort required for staff members to learn and navigate the system, thereby reducing resistance to adoption and encouraging consistent system utilization. The system also incorporated clear instructions, tooltips, and

navigational cues that helped users understand and make full use of its features, further reinforcing their confidence and comfort in using the platform.

Moreover, applying the TAM framework ensured that the proposed system was not only technically functional but also embraced and regularly used by its end users, as a system perceived as useful and easy to use was more likely to deliver sustained operational benefits to Gales Urban Haven. Ultimately, by anchoring the design and development process within the theoretical principles of the Technology Acceptance Model, the researchers ensured that the Condominium Booking and Operations Management System was a practical, user-centered solution that effectively supported the operational goals and digital transformation of Gales Urban Haven under One Spatial Iloilo.

The CRUD Operational Model

The foundational database operations model, Create, Read, Update, Delete (CRUD), served as the structural template for all data management functionalities incorporated in the proposed system. Every entity managed by the system, whether tenant records, booking entries, payment transactions, or maintenance requests, was governed by a complete set of CRUD operations, ensuring that authorized users were able to create new records, retrieve existing ones, modify information as circumstances changed, and remove outdated entries in a controlled and auditable manner. Furthermore, the CRUD model provided a consistent and standardized approach to data management across all system modules, ensuring that every functional component interacted with the centralized database in a uniform, reliable, and predictable manner.

The structured application of CRUD operations also contributed to the overall integrity and accuracy of the system's data, as each operation was subject to validation rules and access controls that prevented unauthorized or erroneous modifications to critical business records. Additionally, the CRUD model served as the foundation for the system's audit trail functionality, as every create, update, and delete action performed within the system was recorded and timestamped, allowing administrators to trace the history of any record and maintain full accountability over all data management activities.

The read operations embedded within the CRUD framework also supported the system's reporting and data retrieval functions, enabling administrators and business owners to access accurate and up-to-date information quickly and efficiently whenever it was needed for operational or decision-making purposes. Moreover, adopting the CRUD model simplified the development and maintenance of the proposed system's database layer, as the well-established and widely recognized nature of the framework provided the development team with a clear and structured blueprint for implementing all data management functionalities. Ultimately, the CRUD operational model ensured that the Condominium Booking and Operations Management System maintained a high standard of data organization, accuracy, and security, providing Gales

Urban Haven with a robust and dependable platform for managing all aspects of its condominium rental operations.

Direct Influence of the Literature on the Proposed System's Design

The reviewed literature not only validated the need for the proposed system but also exerted a direct and traceable influence on its specific design features, functional modules, and technical architecture. The mapping of key literature-derived insights to the corresponding design decisions reflected the features and functionalities incorporated into the Condominium Booking and Operations Management System.

Validation of the Proposed System's Relevance and Necessity

The weight of evidence accumulated across the reviewed literature collectively validated the relevance, necessity, and appropriateness of the Condominium Booking and Operations Management System proposed in this study. This validation operated at multiple levels. At the problem level, the literature confirmed that the operational challenges documented at Gales Urban Haven were well-recognized phenomena with known root causes and known solutions. At the solution level, the literature established that technology-enabled, integrated property management systems represented the most effective and sustainable response to these challenges. At the design level, the literature provided specific, evidence-based guidance on the features, architecture, and development approach that maximized the probability of the system achieving its intended outcomes.

Furthermore, the literature affirmed that the investment in such a system yielded benefits that extended well beyond operational convenience. Organizations that successfully implemented integrated property management systems were positioned to offer a higher standard of service to tenants, operate with greater financial discipline and transparency, respond more effectively to regulatory oversight, and scale their operations with confidence as their property portfolios grew. For Gales Urban Haven, these outcomes translated directly into enhanced competitiveness within the Iloilo City condominium market, improved tenant retention, and a stronger operational foundation for sustainable long-term growth.

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CHAPTER II

METHODOLOGY

3.1 Research Design

This study employed a developmental research design that covered the planning, design, development, implementation, and evaluation of a web-based Condominium Booking and Operations Management System. The design was appropriate because it allowed the researchers to build a functional system that addressed problems in manual condominium and rental management, and to evaluate its functionality, usability, reliability, and effectiveness [1].

The system targeted common problems in manual operations: double bookings, misplaced records, delayed transactions, inaccurate computations, data redundancy, and inefficient report generation. The study proceeded through requirements gathering, system analysis, system design, development, testing, evaluation, and implementation. During requirements gathering, the researchers collected information on current management practices, challenges, and user needs from the target stakeholders. This data served as the basis for the system's features.

In the design and development stage, the researchers created the application's structure, interface, database, and functions for managing customer information, room reservations, booking schedules, billing, payment monitoring, and report generation. User authentication and security features were included to protect the confidentiality and integrity of stored data.

After development, the system underwent functionality, usability, reliability, and user acceptance testing to identify errors, evaluate performance, and confirm that it met user requirements. Evaluator feedback was used to refine the system.

3.2 System Development Methodology

The Condominium Booking and Operations Management System for Gales Urban Haven (operating as One Spatial Iloilo) was developed using the Agile Software Development Model. Agile is an iterative approach that emphasizes flexibility, collaboration, continuous feedback, and incremental delivery [2]. Unlike the Waterfall Model, which requires complete requirements definition before development begins [3], Agile accommodates changing requirements. This suited the project because it integrates several modules: booking management, payment tracking, revenue reporting, calendar-based scheduling, and maintenance monitoring.

1) Requirements Analysis: The team examined the existing manual processes for recording reservations, tracking payments, monitoring maintenance, and scheduling. Methods included consultations with property administrators and key personnel, structured interviews, observation of daily workflows, and review of existing records. The findings were translated into functional requirements (what the system must do) and non-functional requirements (performance, security, usability, reliability). Stakeholders, user roles, access levels, and technical constraints were also identified. The resulting requirements specification guided all later phases.

2) System Design: The requirements were translated into an architectural blueprint covering system architecture, database schema, interface layouts, navigation flows, and module interdependencies. The design followed principles of clarity, simplicity, security, and scalability. Wireframes and mockups were reviewed to validate the interface before coding began. Entity-relationship diagrams (ERDs) and data flow diagrams (DFDs) defined the centralized database and data movement. Security measures, including authentication, access control, and data encryption, were incorporated at this stage.

3) System Development: The modules were implemented in iterative cycles: booking management, payment tracking, revenue reporting, calendar-based scheduling, maintenance monitoring, the centralized database, and user authentication. Each module was built, integrated, and reviewed in turn. The team followed modular code design, consistent coding standards, code documentation, and version control. The modules were integrated so that data flowed between them. For example, reservations recorded in the booking module automatically updated unit availability in the calendar module, and payments recorded in the payment module were reflected in the revenue reporting module.

4) Testing: Testing was conducted at multiple levels:

- Unit testing verified individual modules and functions under normal and boundary conditions.
- Integration testing verified module interaction and data transfer.
- System testing evaluated the integrated system's responsiveness, data accuracy, error handling, security, and accessibility.
- Performance testing assessed stability under varying loads.
- User acceptance testing (UAT) had selected administrators and end users provide structured feedback on usability and suitability.

- Defects were documented, prioritized, and resolved before deployment.

5) Deployment: The system was deployed as a functional prototype and presented to the administrators and key personnel of Gales Urban Haven for demonstration and evaluation. The team configured the system, created and assigned user accounts, and initialized the database. Orientation and walkthroughs familiarized administrators with the system. Feedback from this phase was used to identify remaining refinements, consistent with Agile's continuous improvement cycle.

3.3 System Architecture

The system was built on a client-server architecture, a standard paradigm for web-based information systems [4]. It separates the client tier (presentation and user input) from the server tier (request processing, business logic, and data management).

This architecture was chosen for three reasons. First, multiple authorized users can access the system simultaneously from different devices through a standard web browser, with no client-side installation. Second, centralized data processing and storage ensure that all users work with the same current data. Third, updates to application logic or the database are applied only on the server, which simplifies maintenance. These characteristics suit One Spatial Iloilo, where administrative staff use different workstations within the property management office.

1) Client Tier (Presentation Layer): The client tier is the front end through which administrators and property managers interact with the system. It runs in a web browser and is built with HyperText Markup Language (HTML) for structure, Cascading Style Sheets (CSS) for presentation, and JavaScript for dynamic behavior. The interface uses organized menus, dashboards, tables, forms, summary cards, and interactive calendar views to give access to the booking, payment, revenue, scheduling, and maintenance modules. It is responsive and works on desktop, laptop, and tablet screens. User actions, such as submitting a booking, recording a payment, or updating a maintenance request, are sent to the server as structured requests. The client then displays the server's response as immediate feedback.

2) Server Tier (Application and Logic Layer): The server tier is the central processing engine. It receives requests from the client, executes the corresponding business logic and data operations, and returns responses. Its functions include validating booking submissions, computing payment balances and revenue, enforcing scheduling rules and conflict prevention, managing maintenance records and status updates, and generating financial and operational

reports. The server-side application was developed using [framework/language, e.g., Laravel/PHP, Node.js/Express], which handles concurrent user requests efficiently [5].

3) Data Tier (Database Layer): All operational data are stored in a centralized [DBMS, e.g., MySQL] database accessed only through the server tier, ensuring data integrity and security.

4) Data Flow and Component Integration: The three tiers operate as one integrated system. When an administrator submits a booking, the client sends a request to the server through the defined communication protocol. The server validates the data against business rules, checks unit availability in the database, records the booking, and returns a confirmation. The client then updates the booking dashboard and calendar view.

Payments follow the same path. On recording a payment, the server computes the updated outstanding balance, stores the transaction, and propagates the new figures to the revenue monitoring module and the tenant's payment history in real time. This cross-module propagation keeps every section of the system consistent and avoids the data inconsistencies and lags common in manual record-keeping. The architecture also allows new modules, features, or user roles to be added without disrupting existing functionality.

3.4 Tools and Technologies

The system uses a full-stack JavaScript/TypeScript architecture, chosen for type safety, developer productivity, and maintainability. Each tool was selected for its maturity, community support, performance, and fit with the project's requirements.

3.4.1 Frontend Technologies

React 19: An open-source, component-based UI library maintained by Meta [6]. Its reusable components manage their own state, and its concurrent rendering model keeps the interface responsive under heavy load. Its declarative paradigm simplifies reasoning about state and supports unit testing of individual components.

TypeScript: A statically typed superset of JavaScript developed by Microsoft [7]. Compile-time type checking reduces runtime errors, and shared type definitions act as contracts between frontend and backend.

Tailwind CSS: A utility-first CSS framework [8]. Unused styles are purged in production builds, and its responsive utilities support a mobile-first layout. Its design tokens (colors, spacing, typography, breakpoints) were customized to the project's visual specifications.

Vite: A build tool and development server [9]. It uses native ES Modules for fast startup and hot module replacement (HMR) in development, and Rollup for production builds with code splitting and tree shaking. It also handled path aliasing, environment variables, and API proxying.

3.4.2 Backend Technologies

Node.js: A cross-platform JavaScript runtime built on the V8 engine [10]. Its event-driven, non-blocking I/O model handles many concurrent requests efficiently, and using one language across the stack allows code and types to be shared. The npm ecosystem supplied common modules for authentication, validation, and API communication.

Express: A minimal web framework for Node.js [11] used to build the RESTful API. Its middleware pipeline handles request logging, JSON Web Token (JWT) authentication and authorization, input validation, error handling, and Cross-Origin Resource Sharing (CORS). Endpoints use standard HTTP methods (GET, POST, PUT, PATCH, DELETE) and return JSON.

TypeScript (backend): Strongly typed request and response schemas were defined with TypeScript interfaces and the runtime validation library Zod [12]. Combining compile-time and runtime validation ensures the server processes only well-formed data.

tsx and esbuild: tsx runs TypeScript directly during development, using esbuild for fast transpilation [13]. esbuild also bundles the production backend into a single deployable JavaScript file, avoiding complex Webpack or Babel configurations.

3) Database: MySQL is the relational database management system [14]. It stores tenant profiles, bookings, payments, unit availability, and maintenance logs. Primary keys, foreign key constraints, and normalization enforce integrity, while indexing and transaction support enable efficient real-time reporting and search.

4**) Development Environment**: Visual Studio Code was the integrated development environment (IDE) for all phases. Its extensions provided linting, formatting, database clients, and debugging, and its built-in Git integration supported version control and collaboration.

TABLE I. SUMMARY OF TECHNOLOGIES

Layer	Technology	Purpose
Frontend	React 19	Component-based UI
Frontend	TypeScript	Type-safe programming
Frontend	Tailwind CSS	Utility-first styling

Frontend	Vite	Dev server and production bundler
Backend	Node.js	Server runtime
Backend	Express	RESTful API framework
Backend	TypeScript, Zod	Type safety and runtime validation
Backend	tsx, esbuild	Development execution and production bundling
Database	MySQL	Relational data storage
Tooling	Visual Studio Code, Git	IDE and version control

5) **Rationale for Technology Selection:** The stack was chosen for four reasons:

- Unified language: TypeScript on both tiers reduces code duplication and mismatches between client and server data models.
- Performance: Vite and esbuild shorten development feedback loops and build times.
- Community support: All selected tools have wide adoption, documentation, and maintained packages.
- Scalability: React's component model, Express's modular middleware, and TypeScript's type checking allow incremental growth with a lower risk of regressions.

3.5 Data Gathering Techniques

Two techniques were used to understand the operating environment at Gales Urban Haven (One Spatial Iloilo) and to ground the functional requirements in real practice.

1) Process Analysis: The researchers examined the existing manual workflows through direct observation of daily administrative activities, structured interviews with staff who manage bookings and financial records, and review of the paper-based and spreadsheet tools in use. Each process was mapped end to end, from tenant inquiry through booking confirmation, payment collection, maintenance request, and unit handover. This exposed three main problems: transcription errors from manual data transfer, booking conflicts caused by the lack of a centralized schedule, and forgotten maintenance concerns caused by informal tracking. The findings shaped the functional specifications of the core modules.

2) Review of Existing Platforms: The researchers reviewed property management systems used in residential and hospitality contexts. They examined how these systems handle booking, payment tracking, calendar scheduling, maintenance monitoring, and reporting. The systems were evaluated against the needs of a small-to-medium condominium rental business, which identified applicable best practices and features that were over-complex for this context. Interface conventions, data models, reporting formats, and security approaches informed the design of comparable components.

3.6 Testing Procedures

Testing was conducted in three sequential phases.

1) Unit Testing: Individual modules were tested in isolation: authentication and login, booking creation and retrieval, payment recording and calculation, calendar availability updates, and maintenance request handling. Test cases specified input conditions, expected outputs, and actual outputs, covering valid, invalid, and boundary inputs. The team wrote and ran the tests during implementation, following a test-driven approach where feasible. Discrepancies were traced to their root causes and corrected before the module advanced.

2) System Testing: The integrated system was tested as a whole in an environment mirroring the intended production configuration. Testing covered:

- Integration checks: a booking appears correctly in the calendar, payments aggregate in the revenue dashboard, and maintenance requests flag the correct unit status.
- Performance testing: responsiveness under realistic concurrent usage.
- Security testing: authentication controls and role-based access on protected endpoints.
- Regression testing: confirming that defect fixes did not break verified functions.

3) User Acceptance Testing (UAT): Administrative staff and property management personnel of Gales Urban Haven performed realistic scenarios: recording a booking, processing a payment and verifying the revenue summary, checking unit availability, submitting and updating a maintenance request, and generating a financial report for a given period. Participants completed the tasks independently and gave structured feedback on interface clarity, navigation, and output accuracy. The feedback informed final refinements to the interface, workflows, and documentation before deployment.

3.7 Evaluation Criteria

The system was evaluated on five criteria aligned with the research objectives, using quantitative measurements, structured user feedback, and technical verification. Evaluators included administrators, business owners, and other authorized end users. The results were documented in an evaluation report.

1) **Functionality**: This criterion measures whether the system completely and accurately performs all specified tasks: reservation records, payment processing, revenue and occupancy reports, calendar availability, and maintenance logging. Test cases based on actual Gales Urban Haven workflows were run under standard and boundary conditions, with pass rates recorded. The system is functionally adequate when all defined use cases run without data errors, unhandled exceptions, or failures, and when the modules exchange data without loss or synchronization errors.

2) **Usability**: This criterion measures how easily staff learn the system, navigate between modules, and complete tasks without extensive training. It was assessed through structured observation and feedback during UAT, plus a post-UAT satisfaction survey. The results identified interface elements or navigation flows needing simplification.

3) **Accuracy**: This criterion measures the correctness of computations and records, especially payment totals, outstanding balances, monthly revenue, and occupancy rates. System outputs were compared against manually verified benchmark values from known datasets and checked for reproducibility across repeated runs. Accuracy also covers the integrity of database linkages between bookings, tenants, units, and payments.

4) **Security**: This criterion measures protection of operational and financial data. The evaluation verified the following:

Unauthenticated requests to protected endpoints are rejected.

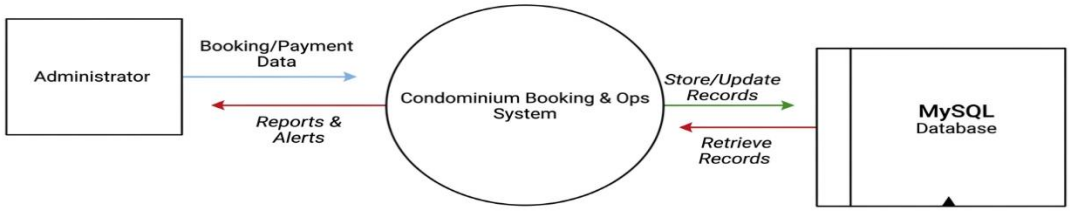
- Expired or forged JWTs are refused.
- Role-based access control restricts users to their permitted scope.
- Parameterized queries prevent SQL injection.
- HTTPS is enforced.
- Credentials are stored with industry-standard hashing.

5) **Efficiency**: This criterion measures the reduction in time, manual effort, and error compared with the manual methods. Staff task times were compared against baseline estimates for equivalent manual tasks. The indicators were booking confirmation time, elimination of manual payment reconciliation, prevention of double bookings through automated availability checks, and maintenance resolution time. Reduced manual workload was the primary indicator of improvement.

3.8 Supporting Diagrams

The following diagrams document the system's data flow and data structure. They were built to standard notation and served as design references for the development team, academic reviewers, and future researchers.

1) Data Flow Diagram (DFD), Level 0: The Level 0 DFD gives the context of the whole system (Fig. 3). The Administrator supplies booking records, payment data, maintenance requests, and scheduling information. The system validates and processes these inputs under its business rules and stores them in the MySQL database. It also retrieves stored records to generate reports, populate dashboards, and return operational summaries to the Administrator. This two-way flow keeps data consistent and available in real
time.

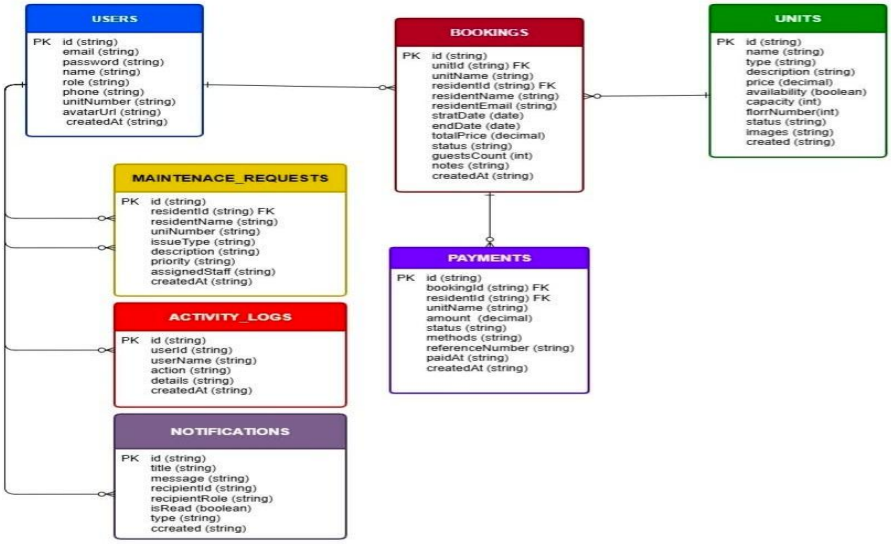


2) Entity Relationship Diagram (ERD): The ERD defines the MySQL schema, organized around five entities (Fig. 4):

- Users: credentials, profile data, and role designations governing access.
- Units: unit type, floor level, monthly rate, and availability status.
- Bookings: the central transactional entity, linking tenants and units through foreign keys to Users and Units, with check-in and check-out dates, status, total rental amount, and notes.

- Payments: an auditable ledger of amounts, dates, and payment methods, linked to Bookings by foreign key.
- Maintenance: issue descriptions, priority, status, and reported and resolved timestamps, linked to specific units.

The one-to-many relationships enforce referential integrity and support cross-module reporting and
queries.



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CHAPTER IV

4.2 Overview of the Developed System

The system developed for this study, titled One Spatial Iloilo: Web-Based Condominium Booking and Operations Management System, is a centralized web-based platform engineered to streamline booking, payment processing, and administrative workflows for One Spatial Iloilo [1]. Prior to system deployment, core operational tasks—such as unit reservations, payment tracking, maintenance requests, and occupancy monitoring—relied on fragmented manual processes and disconnected digital spreadsheets. This operational setup resulted in record consolidation delays, misplaced physical reservation forms, and administrative overhead due to manual cross-referencing across separate ledgers [2].

To resolve these inefficiencies, the developed architecture integrates separate Resident and Administrator portals over a single unified relational database [3]. Data submitted by a user automatically propagates across administrative interfaces in real time, mitigating duplicate entries and maintaining system-wide data integrity.

Core System Modules and Functional Architecture

Access to the platform is managed via an authentication interface hosted on a public-facing landing page displaying property details and amenities. Upon successful authentication, users are routed to their designated portal based on role permissions:

1. Resident Portal: Functions as the primary client interface for tenants and unit owners. The main dashboard displays real-time summaries of active bookings, historical payment ledgers, and outstanding balances. Additional functionalities enable users to:

- Submit and track unit or facility maintenance requests.
- Review transaction histories and profile settings.
- Receive automated operational notifications.

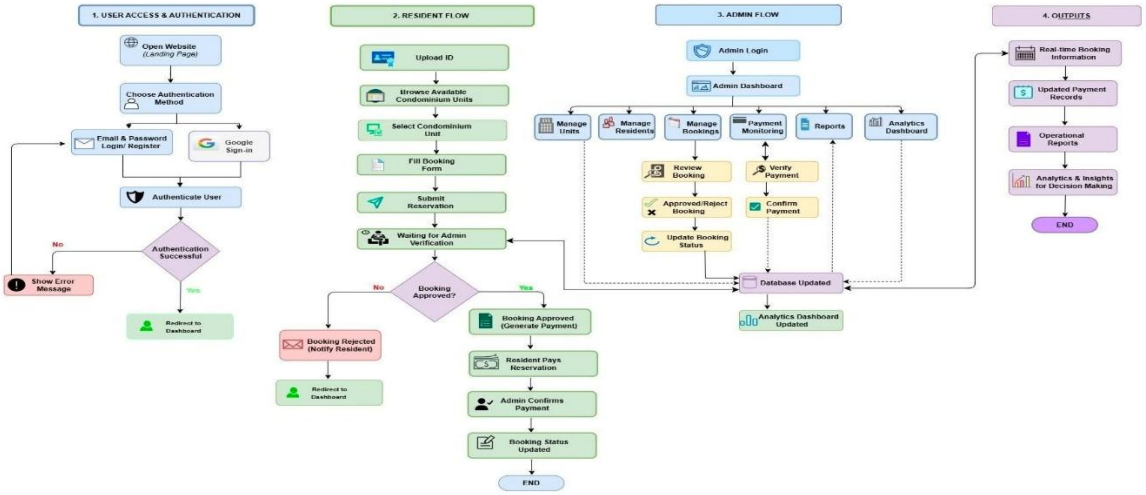
2. Administrator Portal: Designed for property management personnel, the admin interface aggregates operational metrics including total bookings, unit availability, total revenue, and live occupancy rates. Management capability is partitioned into dedicated modules:

- Booking Management Module: Controls reservation state transitions (e.g., Pending, Confirmed, Cancelled, Completed).
- Unit Management Module: Maintains unit inventory, metadata (floor, unit type, pricing structure), and availability statuses.
- Payment Tracking Module: Tracks financial transactions and classifies accounts as Paid, Pending, or Overdue.
- Analytics Dashboard: Generates aggregate reports on revenue trends, booking volumes, and occupancy distribution over selected time horizons.
- Resident Records Module: Manages tenant profiles and unit assignment mapping.
- Maintenance Requests Module: Facilitates the logging, assignment, and status tracking of reported facility issues.
- Notifications and Settings Module: Oversees system-wide broadcast messages and security configurations.

By centralizing data entry, the system eliminates redundant record-keeping, establishes a single source of truth, and optimizes operational efficiency across property management workflows [4].

Figure 4.1 illustrates the overall workflow of the developed

system.



4.3 System Development Methodology

The development of the system followed a structured, phased software development methodology. The lifecycle progressed sequentially through requirements gathering, planning, design, implementation, testing, deployment, and ongoing maintenance [5]. Adopting a phase-gated approach ensured that core structural dependencies—such as database schema design and authentication protocols—were fully validated before subsequent engineering iterations [6].

1) Requirements Gathering and Analysis

Requirements were elicited through stakeholder interviews, informal discussions with residents and administrative staff of One Spatial Iloilo, and an audit of legacy manual workflows. The gathered data was classified into two primary domains:

- Functional Requirements: Defined operational logic including account authentication, unit reservation processing, payment recording, maintenance ticket logging, and real-time report generation (e.g., occupancy and revenue analytics).
- Non-Functional Requirements: Defined quality constraints including user interface (UI) accessibility for non-technical users, system latency targets for dashboard queries, and data protection mechanisms for personally identifiable information (PII) and financial records [7].

These specification criteria were consolidated into a formal Software Requirements Specification (SRS) document to serve as the baseline for system architecture planning.

2) System Planning

- The system architecture was decomposed into discrete functional modules mapped directly to stakeholder requirements. Execution was scheduled based on module dependency: core identity and authentication services were prioritized prior to higher-level transactional features.
- Technological stack selection (front-end interfaces, server-side application logic, and relational database management system) was evaluated against implementation constraints and delivery milestones. Operational risks—such as delayed baseline data collection or scope expansion—were identified, evaluated, and assigned mitigation protocols early in the planning lifecycle.

3) System Design

The system design phase translated specifications into technical blueprints:

- User Interface Design: Wireframes and high-fidelity mockups were generated for all application views (landing page, authentication portals, resident dashboard, admin management modules) to establish a unified visual taxonomy and navigation hierarchy.
- Data Architecture: An Entity-Relationship (ER) model was constructed to define core domain entities (Users, Units, Bookings, Payments, and Maintenance Requests) and their relational cardinality (e.g., mapping a booking record to a unique resident entity and payment ledger entry).
- System Architecture: The multi-tier architecture detailing the interactions between the presentation, application logic, and data storage layers was formalized [8].

4) Software Development

Implementation proceeded iteratively by module in accordance with the project schedule. Each module was constructed through a bottom-up workflow: UI layout development, application logic integration, and database binding. The primary authentication module was developed first to establish Role-Based Access Control (RBAC) across resident and administrator contexts. Subsequent client-side features (dashboard, booking creation, payment tracking, maintenance logs) and administrative control interfaces (unit management, payment verification, analytics engine) were systematically engineered. Source code versioning was maintained throughout the

phase, with code reviews conducted against the baseline interface specifications and data schemas.

5) System Verification and Testing

- Functional validity and stability were verified through a multi-tier testing framework:
- Unit Testing: Validated isolated algorithmic components, such as automated balance calculation functions and dynamic filtering queries.
- Integration Testing: Verified inter-module communication, ensuring that client-side state changes (e.g., a submitted booking request) immediately updated administrative database records.
- System Testing: Executed end-to-end user journeys across complete operational lifecycles (e.g., user registration \rightarrow unit reservation \rightarrow payment submission \rightarrow administrative verification).
- User Acceptance Testing (UAT): Conducted with a sample cohort of residents and staff to identify usability bottlenecks and workflow frictions under simulated operational conditions. Identified defects were logged, corrected, and re-evaluated prior to deployment.

6) Deployment

Following verification, the application was migrated from local development environments to production cloud infrastructure. Production deployment steps included:

- Initializing and configuring the production database server.
- Executing database schema migrations and seeding initial system configurations.
- Deploying front-end assets and server-side runtime services.
- Performing environment validation checks to eliminate discrepancies between local and production runtime behavior.

Upon verification, access credentials were issued to participating residents and staff, marking formal system handover.

7) Maintenance

Post-deployment operations focus on continuous system maintenance, including application performance monitoring, bug remediation, security patch distribution, and framework updates. Furthermore, incremental refinements based on post-launch administrative feedback (e.g., report formatting adjustments and table filter optimizations) are systematically implemented to ensure ongoing operational reliability [9].

8) Agile Development Framework

In addition to the phase-gated engineering pipeline, the overall development cycle can be framed within an Agile development paradigm. Under this framework, requirements are decomposed into iterative sprint cycles, enabling continuous system delivery, feedback integration, and continuous feature refinement based on empirical user feedback.

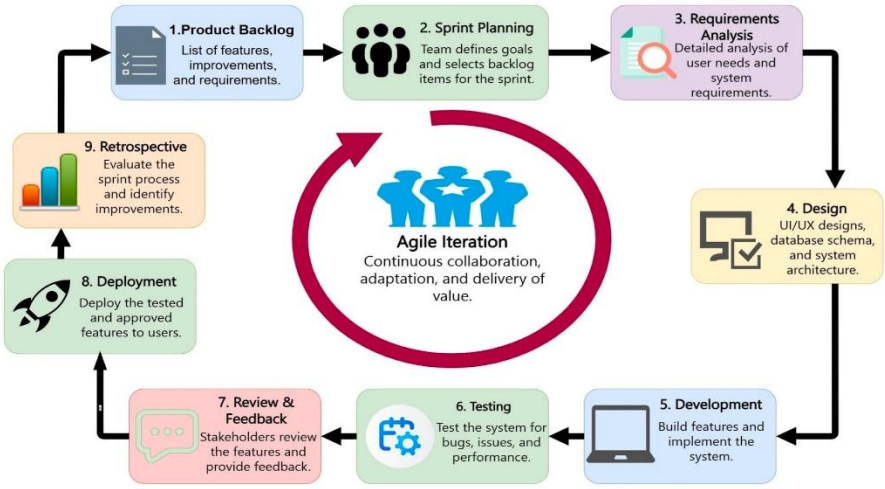


Figure 4.2 presents the Agile Software Development Life Cycle adopted in the study.

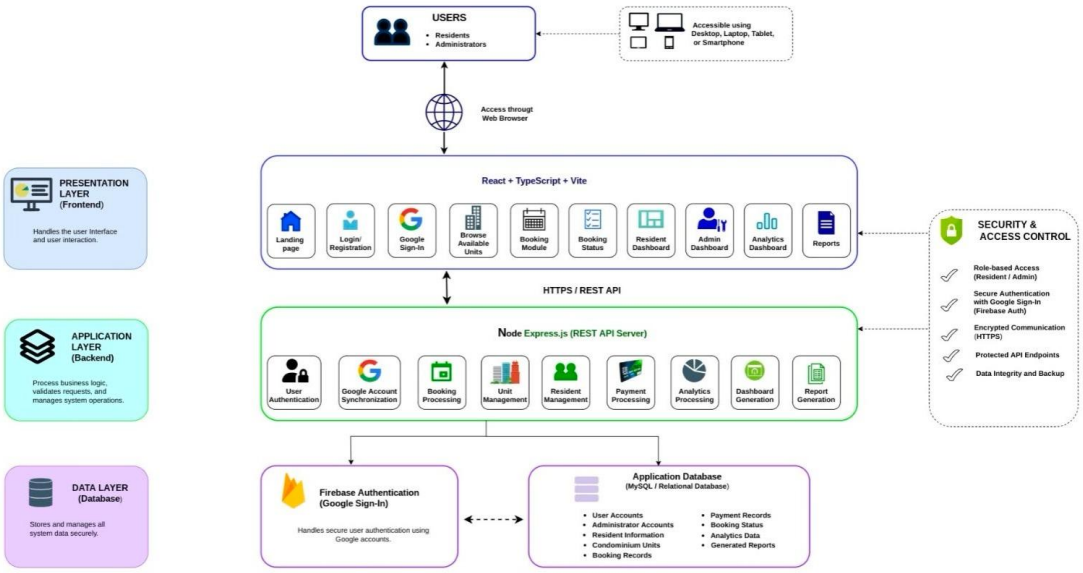
9) Agile Process Framework

Under the Agile development approach, the engineering process was organized into iterative sprint cycles comprising requirements specification, sprint planning, incremental feature development, continuous testing, and stakeholder feedback integration [10]. This iterative

paradigm is particularly suited for condominium management systems, as it permits continuous refinement of transactional workflows—such as reservation scheduling, payment verification, and maintenance tracking—based on practical operational feedback [11].

4.4 System Architecture

The system was engineered around a traditional Three-Tier Architecture, decoupling the presentation, application logic, and data storage layers [12]. This architectural separation ensures modular maintainability: updates to presentation layouts operate independently of business logic, while database schema modifications do not disrupt core processing routines.



1) Presentation Layer

The presentation layer serves as the primary client-facing tier, rendering data obtained from lower application tiers into responsive graphical user interfaces (GUIs). It captures user inputs (e.g., booking requests or profile modifications) and transmits them to the application layer for validation and execution. Interfaces were implemented with responsive web design patterns to ensure cross-device compatibility across desktop and mobile browsers. Key presentation modules include:

- Landing Page: Displays property overview and public amenities.

- Authentication Page: Handles user login and identity submission.
- Resident Dashboard: Provides client-specific financial and booking overviews.
- Administrator Dashboard: Aggregates macro-level operational and performance metrics.
- Booking Management Interface: Facilitates reservation processing.
- Unit Management Interface: Displays inventory and rate configurations.
- Payment Tracking Interface: Logs financial records and transaction statuses.
- Analytics Dashboard: Visualizes revenue trends and occupancy statistics.
- Resident Profile Interface: Enables account information management.
- Maintenance Requests Interface: Facilitates ticket submission and monitoring.
- Notification Interface: Displays system alerts and administrative broadcasts.

2) Application Layer

Operating between the presentation and data tiers, the application layer encapsulates core business rules, computation engines, and transactional workflows [13]. It enforces Role-Based Access Control (RBAC), validates operational constraints (e.g., preventing double-booking of active units), calculates dynamic variables (e.g., outstanding balances and occupancy percentages), and coordinates inter-tier communications.

Core functionalities include:

- Authentication and Session Management Engine: Handles identity verification and access permissions.
- Booking Execution Logic: Validates unit availability and processes reservation states.
- Payment Processing Engine: Handles transaction validation and balance calculations.
- Maintenance Workflow Controller: Manages ticket creation, status progression, and task assignment.
- Notification Engine: Generates and routes real-time system alerts.
- Analytics and Reporting Engine: Aggregates transactional data into statistical metrics.
- Profile Management Logic: Processes account updates and updates user attributes.

3) Data Layer

The data layer manages persistent storage and retrieval of relational system records. Constructed in accordance with the Entity-Relationship (ER) model established during the system design phase, entity cardinalities and relational constraints are enforced at the database schema level to uphold transactional integrity (ACID properties) [14].

Stored entities encompass:

- User Account Records: System credentials, roles, and personal profiles.
- Unit Records: Physical metadata, pricing tiers, and real-time availability states.
- Booking Records: Reservation details, time spans, and status identifiers.
- Payment Records: Transaction ledgers, payment receipts, and balance statuses.
- Maintenance Request Records: Facility issue logs, repair statuses, and timestamps.
- Notification Records: System alerts and user messaging archives.

To illustrate multi-tier interactions: when a user executes a booking through the Presentation Layer, the request is forwarded to the Application Layer to verify unit availability and calculate applicable fees. Upon successful validation, the Data Layer writes the transaction record and updates unit status flags. Finally, the Presentation Layer dynamically reflects the updated operational state across both Resident and Administrator portals without requiring manual synchronization [15].

4.5 Hardware and Software Specifications

To ensure optimized system throughput, deployment stability, and client accessibility, system resource requirements were defined for development, production hosting, and end-user execution contexts.

- Requirements: Client-side hardware criteria mandate standard computing or mobile hardware capable of running modern standard web browsers. Server-side hosting hardware requires sufficient processing power and memory allocation to process simultaneous HTTP/HTTPS requests, run database query engines, and generate real-time analytics graphs without latency spikes.
- Software Requirements: The software pipeline comprises specialized frontend and backend frameworks, a Relational Database Management System (RDBMS), authentication libraries, package management utilities, version control infrastructure, and deployment runtime environments.

4.5.1 Hardware Requirements

The minimum and recommended hardware configurations required to support development, administrative operations, and client access for the system are outlined in Table I. Meeting or exceeding these specifications ensures optimal browser rendering, efficient query execution, low-latency dashboard visualization, and robust performance during concurrent user sessions [16].

Table 4.1 Hardware Requirements

Component	**Minimum Specification**	**Recommended Specification**
Processor	Intel Core i3 (8th Generation) or AMD Ryzen 3	Intel Core i5 (10th Generation) or higher / AMD Ryzen 5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	256 GB SSD	512 GB SSD or higher
Monitor	1366 x 768 Resolution	Full HD (1920 × 1080) Display
Internet Connection	10 Mbps	50 Mbps or higher
Mobile Device	Android 10 or iOS 14	Android 13 or iOS 17 or later

Operating the system on or above the recommended hardware configuration guarantees sufficient computational throughput to handle resource-intensive web client operations—such as multi-chart data visualization, interactive payment ledger queries, and real-time report compilation [17].

Furthermore, higher network bandwidth and memory capacity minimize page load latency and optimize application responsiveness across desktop and mobile execution environments.

4.5.2 Software Requirements

The software software stack and development utilities utilized to build, test, and execute the system are detailed in Table II. Software choices were evaluated based on framework stability, modularity, active community support, and performance suitability for decoupled modern web architectures [18].

Table 4.2 Software Requirements

Component	**Purpose**
Windows 10/ Windows 11	Operating System
Visual Studio Code	Source Code Editor
Node.js	JavaScript runtime environment
npm	Dependency and package management
React	Front-end Development Library
Typescript	Front-end programming language
Vite	Front-end build tool and Development server
Express.js	Backend development framework
Firebase Authentication	Google sign in authentication
JSON database	Local data storage

Google Chrome	Web browser for testing and system access
Git	Version control
Github	Source code repository

The client interface was engineered using React with TypeScript and built via Vite to maintain high performance, strict type safety, and fast DOM re-rendering [19]. Server-side REST APIs and business logic execution were hosted on a Node.js environment using Express.js. Account identity verification was integrated through Firebase Authentication to enable secure OAuth 2.0 Google Sign-In protocols. System state persistence during initial development iterations was managed via a structured JSON dataset (db.json). Version control, code review pipelines, and source integrity were maintained using Git and hosted on GitHub repositories [20].

4.6 Database Design

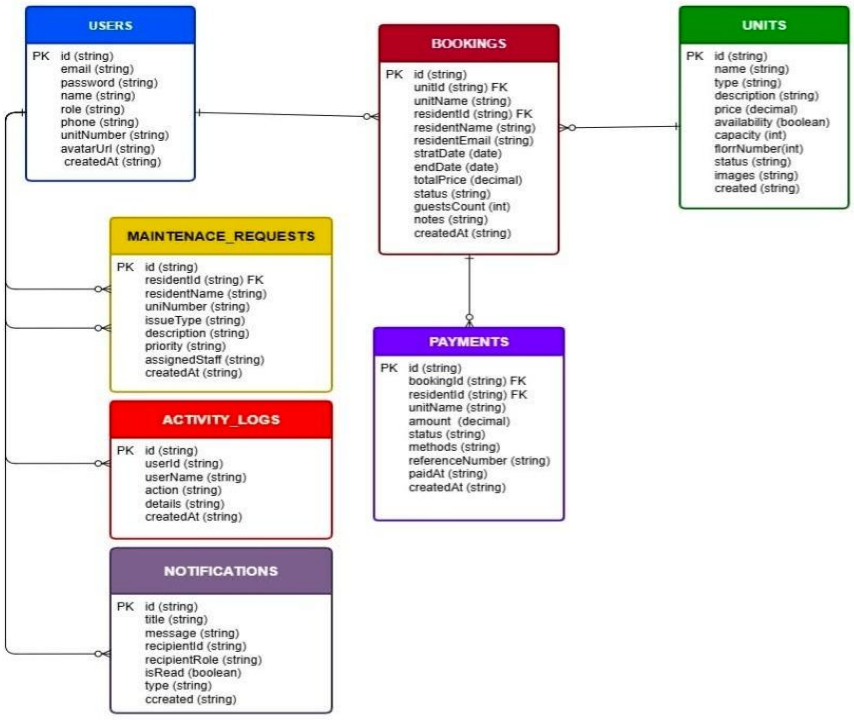
The data layer of the application was constructed using a normalized relational database schema comprising seven interconnected tables. The design applies Boyce-Codd Normal Form (BCNF) principles to eliminate data redundancy while maintaining referential integrity across operational transactions [21].

The database entities and their relational mappings are structured as follows:

- users Table: Serves as the central entity table storing user accounts, authentication credentials, profile data, and access roles (administrator or resident). The primary key id is referenced as a foreign key across multiple downstream tables.
- units Table: Holds property metadata including unit capacity, floor allocation, rental rates, and real-time availability states. It maintains an independent existence prior to reservation assignment.
- bookings Table: Establishes a many-to-one relationship (1:N) with both the units table via unitId and the users table via residentId. This binds a specific resident to a unique unit over a defined reservation period.

- payments Table: Linked to the bookings table via bookingId (1:N) and the users table via residentId. This structure allows transaction logs to be audited against their underlying reservation entries and resident billing statements.
- maintenance_requests Table: Maintains a foreign key link (residentId) to the users table, ensuring facility maintenance issues are logged, assigned, and tracked per resident account.
- notifications Table: Supports targeted messaging via recipientId (referencing users.id) and role-based broadcasting via recipientRole (targeting administrator or resident groups).
- activity_logs Table: Captures system-wide audit trails tied to userId. To optimize query performance during administrative log auditing, selective denormalization is applied by retaining userName and userRole directly within the log entries, avoiding expensive join operations across high-volume audit tables [22].

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.



The primary entities of the database include:

- Users
- Bookings
- Units
- Payments
- maintenance_requests
- Notifications
- Activity_logs

Users Table

This table stores account credentials and user information.

Field	**Description**
user_id	Primary Key
Email	User email
password	User password
name	Full name
role	Admin/residents
phone	Contact number
Unit_number	Resident unit number
AvatarUrl	Profile image URL
CreatAt	Date Created

Units Table

Stores condominium unit information and availability.

Field	**Description**

id	Primary key
name	Unit name
type	Unit type
description	Unit description
price	Rental price
availability	Availability status
capacity	Maximum guest capacity
floorNumber	Floor number
status	Available, Booked, or Maintenance
images	Unit image URLs
createdAt	Date created

Booking table

Stores resident booking and reservation details.

Field	**Description**
id	Primary key
unitId	Foreign key to units
unitName	Unit name
residentId	Foreign key to users
residentName	Resident name
residentEmail	Resident email
startDate	Booking start date
endDate	Booking end date
totalPrice	Total booking price
status.	Booking status
guestsCount	Number of guests
notes	Additional notes

createdAt	Date created

Payments Table

Stores payment and transaction details.

Field	Description
id	Primary key
bookingId	Foreign key to bookings
residentId	Foreign key to users
residentName	Resident name
unitName	Unit name
amount	Payment amount
status	Payment status
method	Payment method
referenceNumber	Payment reference number
paidAt	Payment date
createdAt	Date created

maintenance_requests table

Stores resident maintenance requests and their status.

Field	Description
id	Primary key
residentId	Foreign key to users
residentName	Resident name
unitNumber	Unit number
issueType	Type of issue

description	Issue description
priority	Priority level
status	Maintenance status
assignedStaff	Assigned staff
createdAt	Date created

notifications table

Stores notifications sent to users.

Field	Description
id	Primary key
title	Notification title
message	Notification content
recipientId	Recipient user ID or all
recipientRole	Recipient role
isRead	Read status
type	Notification type
createdAt	Date created

activity_logs table

Stores records of user activities in the system.

Field	Description
id	Primary key
userId	Foreign key to users
userName	User name
userRole	Admin or resident
action	Performed action

details	Action details
createdAt	Date created

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CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 **Introduction**

This chapter synthesizes the outcomes, conclusions, and operational recommendations derived from the engineering and evaluation of the One Spatial Iloilo: Web-Based Condominium Booking and Operations Management System. System performance was validated through functional verification, user acceptance testing (UAT), and data analysis to determine whether the core objectives were achieved [1]. The developed platform addresses recurring operational bottlenecks—such as amenity scheduling conflicts, manual payment reconciliation, and uncoordinated maintenance tracking—traditionally managed via physical logbooks and disconnected spreadsheets [2].

5.2 Summary of Findings

Evaluation data were gathered from key stakeholders, including condominium administrators, front desk staff, and residents. The empirical evaluation incorporated functional verification,

usability assessments, and standardized software quality evaluation frameworks aligned with ISO/IEC 25010 metrics [3].

- Key findings are summarized as follows:
- Amenity Booking Automation: Automated scheduling for shared facilities (function halls, swimming pools, fitness centers) eliminated reservation collisions. Real-time calendar synchronization and dynamic conflict-checking algorithms mitigated double-booking instances previously common under manual logbook workflows.
- Financial and Billing Operations: Integration of the payment tracking module accelerated dues collection and ledger reconciliation. Administrative reporting timelines were significantly reduced compared to manual receipt processing [4].
- Maintenance Workflow Visibility: Transparent ticket creation and status tracking allowed residents to log unit/facility issues directly, while maintenance personnel updated resolution statuses in real time.
- Usability and System Performance: Standardized usability metrics indicated high interface accessibility across varied technical backgrounds. System latency and load handling remained stable during peak user traffic scenarios.
- Operational Feedback: Minor latency in notification dispatching and requests for expanded onboarding documentation were identified as secondary areas for optimization.

5.3 Conclusion

- Based on the empirical findings, the researchers concluded that the developed web platform successfully satisfied all initial system requirements and objectives:
- Centralization: The multi-tier platform effectively replaced fragmented, paper-based administrative workflows with a unified database structure.
- Workload Reduction: Automating core operational workflows minimized human error and administrative overhead, permitting staff to reallocate resources toward higher-value facility management tasks [5].
- Feasibility: Positive stakeholder evaluation confirmed that a digital management ecosystem is both technically viable and operationally superior to traditional property management methodologies.

5.4 Recommendations

To optimize system execution and guide future academic inquiries, recommendations are structured across three primary dimensions:

1) System Enhancements

- Notification System Optimization: Refine background job execution and web-push protocols to eliminate dispatch latency during peak loading conditions.
- Onboarding Module: Integrate step-by-step interactive application tours to support first-time users and non-technical demographics.
- Native Mobile Application: Develop dedicated cross-platform mobile interfaces (iOS/Android) alongside the web platform to broaden accessibility.
- Advanced Analytics Engine: Expand administrative analytics modules to incorporate predictive budgeting, historical revenue forecasting, and trend analysis [6].
- Security Protocol Hardening: Strengthen data encryption standards (at rest and in transit) and implement automated database backup schedules to protect sensitive resident metadata.

2) Future Research Directions

- Artificial Intelligence Integration: Investigate machine learning models to predict facility demand cycles and generate automated preventive maintenance schedules based on historical utilization logs [7].
- Scalability Analysis: Evaluate platform performance across large-scale, multi-tower residential complexes or gated communities with multi-tier administrative governance structures.
- IoT and Access Control Integration: Incorporate biometric authentication, RFID, or computer-vision-based facial recognition for automated gate access and visitor logging [8].
- Longitudinal Deployment Studies: Conduct multi-year longitudinal impact assessments post-deployment to quantify long-term operational cost savings and administrative return on investment (ROI).

3) Implementation Strategies

- Phased Pilot Deployment: Initiate rollouts via a limited pilot phase (e.g., restricted to a single amenity or selected resident cohort) prior to full-scale platform deployment.

- Stakeholder Training Programs: Conduct structured onboarding workshops for property officers and administrative staff to streamline transition management and reduce operational friction.
- Technical Support Infrastructure: Establish dedicated technical help-desk channels during initial transition months to address user queries and system anomalies promptly.
- Parallel Adoption Period: Run the digital platform concurrently with legacy paper workflows during a brief transition window to ensure data continuity before full legacy phase-out.

5.5 Impact of the Study

- The deployment of the system yields direct operational and academic contributions:
- Operational Impact: Residents benefit from transparent, self-service amenity reservations and financial tracking, mitigating administrative friction. Administrators gain automated audit trails, centralized ledgers, and dynamic reporting tools that reduce manual workload [9].
- Contextual Domain Knowledge: The study contributes to property and facility management literature by demonstrating the feasibility of digital transformation in mid-sized residential complexes without requiring cost-prohibitive infrastructure [10].
- Academic Reference: The system architecture, database schema, and evaluation methodologies serve as a design framework for software engineers, students, and property technology researchers.

5.6. Summary

Chapter V synthesized the evaluation outcomes, conclusions, and strategic recommendations for the Condominium Booking and Operations Management System. The empirical findings confirm that transitioning from manual processes to a centralized digital architecture resolves scheduling conflicts, optimizes financial tracking, and improves maintenance coordination. The chapter concluded with actionable recommendations for technical enhancements, implementation strategies, and future research pathways in property technology.

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